

ActiveLLM: Large Language Model-based Active Learning for Textual Few-Shot Scenarios

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Abstract

Active learning is designed to minimize annotation efforts by prioritizing instances that most enhance learning. However, many active learning strategies struggle with a ‘cold-start’ problem, needing substantial initial data to be effective. This limitation reduces their utility in the increasingly relevant few-shot scenarios, where the instance selection has a substantial impact. To address this, we introduce ActiveLLM, a novel active learning approach that leverages Large Language Models such as GPT-4, o1, Llama 3, or Mistral Large for selecting instances. We demonstrate that ActiveLLM significantly enhances the classification performance of BERT classifiers in few-shot scenarios, outperforming traditional active learning methods as well as improving the few-shot learning methods ADAPET, PERFECT, and SetFit. Additionally, ActiveLLM can be extended to non-few-shot scenarios, allowing for iterative selections. In this way, ActiveLLM can even help other active learning strategies to overcome their cold-start problem. Our results suggest that ActiveLLM offers a promising solution for improving model performance across various learning setups.

1 Introduction

The selection of training examples significantly impacts the performance of models. Even Large Language Models (LLMs), such as GPT-3, show high variances based on training example selection (Zhang et al., 2022b). This dependency is even more pronounced in smaller models such as BERT (Zhang et al., 2022b), which are sometimes preferred for their cost and resource efficiency as well as data protection benefits over APIs. In order to combine the advantages of both the lightness of BERT-like models and the capabilities of LLMs, we explore the classical concept of Active

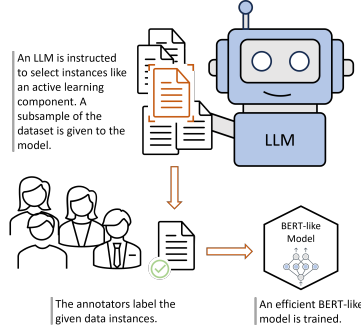


Figure 1: Depiction of ActiveLLM.

Learning (AL) for BERT-like models, extended by LLMs such as GPT-4 (OpenAI, 2023).

AL encompasses strategies integrated into the labeling process to select instances with a high learning impact (Zhang et al., 2022c). Typically, a model is trained iteratively during the annotation process to query instances for labeling by the annotators (oracle). While AL can yield significant learning improvements with mid-sized datasets, it often encounters a ‘cold-start’ problem (Chen et al., 2023), rendering it unsuitable for low-data regimes. The cold-start problem occurs in many AL methods because they lack sufficient data at the start of data labeling to accurately measure informativeness and select informative samples. For example, the uncertainty strategy selects the most uncertain instances based on an iteratively learned classifier during annotation. However, the classifier cannot make accurate uncertainty guesses without enough data.

Furthermore, as most AL strategies require a model to be trained iteratively, the use of pre-trained models such as BERT can result in very high delay times during annotation, making the

ActiveLLM: 基于大语言模型的文本少样本场景主动学习

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Abstract

主动学习旨在通过优先选择最能促进学习的实例来最小化标注工作量。然而，许多主动学习策略都面临“冷启动”问题，需要大量初始数据才能有效。这一局限降低了它们在日益相关的小样本场景中的实用性，而在这些场景中，实例选择具有重大影响。为此，我们提出了 ActiveLLM，一种新颖的主动学习方法，它利用大型语言模型（如 GPT-4、o1、Llama 3 或 Mistral Large）来选择实例。我们证明，ActiveLLM 在小样本场景中显著提升了 BERT 分类器的分类性能，优于传统的主动学习方法，并改进了小样本学习方法 ADAPET、PERFECT 和 SetFit。此外，ActiveLLM 可以扩展到非小样本场景，允许进行迭代选择。通过这种方式，ActiveLLM 甚至可以帮助其他主动学习策略克服其冷启动问题。我们的结果表明，ActiveLLM 为提升各种学习设置下的模型性能提供了一个有前景的解决方案。

1 Introduction

训练样本的选择显著影响模型的性能。即使是大型语言模型（LLMs），例如 GPT-3，其表现也会因训练样本的选择而产生高度差异 (Zhang et al., 2022b)。这种依赖性在 BERT (Zhang et al., 2022b) 等较小模型中更为明显，这些模型有时因其成本与资源效率以及相较于 API 的数据保护优势而更受青睐。为了结合类 BERT 模型的轻量性与 LLMs 的能力优势，我们探索了针对类 BERT 模型的经典主动学习概念，并利用 GPT-4 (OpenAI, 2023) 等

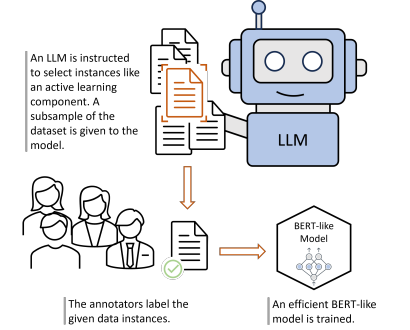


图 1: ActiveLLM 示意图。

LLMs 对其进行了扩展。

主动学习包含集成到标注过程中的策略，旨在选择具有高学习影响力的实例 (Zhang et al., 2022c)。通常，在标注过程中会迭代训练一个模型，以向标注者（或标注方）查询需要标注的实例。虽然主动学习能在中等规模数据集上带来显著的学习改进，但它常遇到“冷启动”问题 (Chen et al., 2023)，使其不适用于低数据量场景。冷启动问题出现在许多主动学习方法中，因为它们的数据标注开始时缺乏足够的信息来准确衡量信息量并选择信息丰富的样本。例如，不确定性策略基于标注过程中迭代学习的分类器来选择最不确定的实例。然而，没有足够的信息，分类器无法做出准确的不确定性猜测。

此外，由于大多数主动学习策略需要迭代训练模型，使用如 BERT 这样的预训练模型可能导致标注过程中产生极高的延迟，使得该过程在实际标注场景中无法使用。在所谓的“模

process unusable in real-world annotation scenarios. In so-called ‘model-mismatch scenarios’, where the instance selection model (query model) differs from the model used for the final application (successor model), AL yields limited gains (Zhang et al., 2022c).

To address these challenges of few-shot and model-mismatch scenarios, we propose ActiveLLM, an AL method leveraging LLMs (see Figure 1). ActiveLLM can select instances with high learning impact even without initial supervised data and requires no training during the annotation process. This method serves both as a standalone AL approach and as a solution to the cold-start problem in other AL strategies. Therefore, our contributions are as follows:

(C1) Novel AL Method: We introduce a novel AL strategy using LLMs such as GPT-4, tailored for few-shot learning to overcome the cold-start problem.

(C2) Efficient and Scalable Methodology: Our approach decouples the query process from the successor model dependency, enhancing scalability and establishing a high degree of practicality compared to conventional AL strategies.

(C3) Rigorous Evaluation: We conduct comprehensive evaluations to demonstrate that ActiveLLM is not constrained to a specific LLM. It exhibits superior performance compared to other AL and few-shot learning methods and can be utilized alongside these methods to achieve beneficial outcomes.

The code for this study will be freely available.

2 Related Work

2.1 Active Learning

The rationale behind AL is that a system selectively chooses instances to be labeled, thereby reducing the labeling effort (Settles, 2012). In this context, annotators serve as oracles, responding to queries from the AL system regarding specific instances. This system utilizes a query strategy that often hinges on measures of uncertainty or diversity. Traditionally, the querying is performed by the machine learning model itself, which undergoes iterative training during the labeling process. For instance, the model might request labels for instances where its uncertainty is highest, using approaches such as Least Confidence (LC) (Cohn et al., 1994), Margin of Confidence (Tong and Koller, 2001), or Prediction Entropy (PE) (Settles

and Craven, 2008). It is then retrained with the newly annotated data.

In addition to uncertainty-based methods, diversity-based approaches also play a critical role in AL. These methods aim to select instances that are diverse to ensure the labeled dataset covers a wide range of feature space. For example, techniques like Embedding KMeans (EKM) (Schröder et al., 2023; Yuan et al., 2020) leverage the model’s ability to embed instances into a latent space. By applying clustering algorithms such as KMeans in this space, the system can identify and select the most diverse examples, thereby enriching the training set with varied and representative data points.

2.1.1 Active Learning with Transformer Models

While AL methods have demonstrated their utility across various traditional machine learning models, their application to transformer models remains challenging. The study by Ein-Dor et al. (2020), among the first to explore AL with pre-trained transformer models such as BERT, shows that AL can enhance the performance of BERT classifiers, particularly in low-data scenarios and when the initial dataset is rich in relevant class instances. However, most of the other works indicate mixed results. For example, Jacobs et al. (2021) combine uncertainty and diversity-based methods, utilizing SentenceBERT embeddings to refine the instance selection process. Their findings indicate that while AL reduces labeling effort, its effectiveness with BERT models is mixed, and the improvements are modest compared to older NLP models. Studies by Yuan et al. (2020); Schröder et al. (2022); Seo et al. (2022); Griebhaber et al. (2020); Houlsby et al. (2011) contribute to a similar mixed picture of AL methods for transformer models. In Table 2.1.3, we present an overview of works utilizing AL for transformer models, categorizing them by query strategies, tested models, and model matching/mismatching scenarios.

Research Gap: The most significant gap in AL research is the use of LLMs, such as GPT-4, to improve AL with BERT-like transformer models, which, to our knowledge, has not been explored previously.

型不匹配场景”中，即实例选择模型（查询模型）与最终应用模型（后续模型）不同时，主动学习带来的增益有限 (Zhang et al., 2022c)。

为了应对少样本和模型不匹配场景的这些挑战，我们提出了ActiveLLM，一种利用 LLMs 的主动学习方法（见图1）。ActiveLLM即使在没有初始监督数据的情况下也能选择具有高学习影响力的实例，并且在标注过程中无需训练。该方法既可作为一种独立的主动学习方法，也可作为其他主动学习策略中冷启动问题的解决方案。因此，我们的贡献如下：

(C1) Novel 主动学习方法：我们引入了一种新颖的主动学习策略，使用 GPT-4 等 LLMs，专为少样本学习设计，以克服冷启动问题。

(C2) Efficient and Scalable Methodology: 我们的方法将查询过程与后续模型的依赖性解耦，与传统主动学习策略相比，增强了可扩展性并确立了高度的实用性。

(C3) Rigorous Evaluation: 我们进行了全面的评估，以证明ActiveLLM不受限于特定的 LLM。与其他主动学习和少样本学习方法相比，它展现出更优越的性能，并且可以与这些方法结合使用以获得有益的结果。

本研究的代码将免费提供。

2 Related Work

2.1 Active Learning

主动学习（AL）的基本原理在于系统有选择地选取需要标注的实例，从而减少标注工作量 (Settles, 2012)。在此背景下，标注者充当了预言机的角色，响应 AL 系统针对特定实例的查询该系统采用的查询策略通常依赖于不确定性或多样性的度量。传统上，查询由机器学习模型自身执行，该模型在标注过程中进行迭代训练。例如，模型可能会请求对其不确定性最高的实例进行标注，采用的方法如最小置信度（LC）(Cohn et al., 1994)、置信度边际 (Tong and Koller, 2001) 或预测熵（PE）(Settles and Craven, 2008)。随后，模型会使用新标注的数据进行重新训练。

除了基于不确定性的方法外，基于多样性的方法在 AL 中也扮演着关键角色。这些方法旨在选择多样化的实例，以确保标注数据集覆盖广泛的特征空间。例如，嵌入 K 均值 (EKM) (Schröder et al., 2023; Yuan et al., 2020) 等技术利用了模型将实例嵌入到潜在空间的能力。通过在该空间中应用 K 均值等聚类算法，系统可以识别并选择最具多样性的样本，从而用多样化和具有代表性的数据点丰富训练集。

2.1.1 Active Learning with Transformer Models

尽管 AL 方法在各种传统机器学习模型中已证明其效用，但其在 Transformer 模型中的应用仍面临挑战。Ein-Dor et al. (2020) 的研究是首批探索将 AL 与预训练 Transformer 模型（如 BERT）结合的工作之一，表明 AL 能够提升 BERT 分类器的性能，尤其是在数据稀缺以及初始数据集富含相关类别实例的场景下。然而，大多数其他研究的结果好坏参半。例如，Jacobs et al. (2021) 结合了基于不确定性和基于多样性的方法，利用 SentenceBERT 嵌入来优化实例选择过程。他们的研究结果表明，虽然 AL 减少了标注工作量，但其在 BERT 模型上的效果不一，且与较旧的 NLP 模型相比，改进幅度有限。Yuan et al. (2020); Schröder et al. (2022); Seo et al. (2022); Griebhaber et al. (2020); Houlsby et al. (2011) 的研究也描绘了 AL 方法在 Transformer 模型上类似的好坏参半的图景。在表??中，我们概述了将 AL 用于 Transformer 模型的相关工作，并按查询策略、测试模型以及模型匹配/不匹配场景进行了分类。

Research Gap: AL 研究中最显著的空白是使用 LLM（如 GPT-4）来改进基于 BERT 类 Transformer 模型的 AL，据我们所知，此前尚未有研究对此进行探索。

2.1.2 Model-Mismatch Scenarios

As transformer models often require substantial computational resources and time (Treviso et al., 2023), the training runtimes for those can be prohibitively long during labeling, making them impractical for real-world applications of AL. Schröder et al. (2022) noted that the incremental training of these models can be so time-consuming, that it can negate the cost savings from reduced labeling effort. Therefore, the scope of AL has been extended to include scenarios where the querying model may differ from the final (successor) model — a phenomenon known as model mismatch — or situations where no model is used for querying.

But again many studies (Baldridge and Osborne, 2004; Schröder et al., 2022; Shelmanov et al., 2021) report that a model mismatch between the query and successor model often leads to unsatisfactory results (Zhang et al., 2022c). While efforts by Shelmanov et al. (2021) and Nguyen et al. (2022) attempt to address this by using smaller, similar models for querying, there remains a dependency on the model type. Similarly, Tsvigun et al. (2022) offer a promising approach for decoupling the query and successor models, although a distilled version is still required in the process.

Research Gap: In contrast to the use of smaller, resource-efficient models for querying, we propose leveraging much larger language models. This approach may seem counterintuitive at first due to concerns about long runtimes, but as these models are capable of zero-shot learning, no training is required (Shliazhko et al., 2024). Furthermore, these models can be accessed in real-time through several free chat interfaces, requiring no machine learning expertise, making them ideal for practical applications.

2.1.3 Cold-Start Problem

While AL can significantly streamline the learning process by reducing labeling efforts, most strategies encounter a cold-start problem. In the absence of sufficient initial labeled data (label seed), the AL system may struggle to make informed predictions about uncertainty or diversity (Yuan et al., 2020). This presents a major challenge for low-data regimes, particularly in few-shot scenarios with very few available data instances. Yuan et al. (2020) propose ALPS, which is one of the only works that directly addresses the cold-start

problem in AL with transformer models. ALPS leverages BERT’s pre-existing masked language modeling objective along with clustering to select instances. The authors hypothesize that this strategy provides more reliable confidence scores in the initial stages of model training than a classifier head. However, the results are mixed which is also later confirmed by Nguyen et al. (2022).

Research Gap: The cold-start problem is a major challenge in AL research, especially for few-shot scenarios. This work aims to overcome this challenge by proposing an AL strategy that needs no initial labeled data, enabling improvements even at the very beginning of the AL process. This allows the method to also be used to address the cold-start problem of other AL methods by providing them with initial data.

2.2 Few-Shot Learning

Few-shot learning refers to methods for scenarios where models need to learn from a limited number of training instances. These scenarios are becoming increasingly relevant due to their importance in real-world problems (Yu and Raschka, 2020) and the increasing capabilities of deep learning models. While LLMs such as GPT-3 inherently have robust few-shot or zero-shot capabilities through in-context learning, our focus is on enhancing few-shot learning in smaller models such as BERT. With specific few-shot learning methods like ADAPET (Tam et al., 2021) these smaller language models can attain similar or even better few-shot performance than GPT-3.

One approach to this is to formulate tasks as cloze-style tests, where certain words in a text instance are missing and the language model is employed to fill in the blanks. This way, no classifier needs to be trained on top of the language model, leading to a more effective utilization of the language models (Gao et al., 2021). Such an approach is exemplified by the ADAPET method (Tam et al., 2021), as well as in the works of Gao et al. (2021); Zhang et al. (2022a); Schick and Schütze (2021a). PERFECT, introduced by Mahabadi et al. (2022), eliminates the need to manually formulate such cloze tasks by incorporating task-specific adapters and specialized label embeddings. Similarly, in SetFit by Tunstall et al. (2022), no cloze tasks are required, as a sentence transformer (Reimers and Gurevych, 2019) is finetuned on the available training data. The instances

2.1.2 Model-Mismatch Scenarios

由于 Transformer 模型通常需要大量的计算资源和时间 (Treviso et al., 2023), 其在标注过程中的训练运行时间可能过长, 以至于在主动学习的实际应用中变得不切实际。Schröder et al. (2022) 指出, 这些模型的增量训练可能非常耗时, 以至于可能抵消因减少标注工作量而节省的成本。因此, 主动学习的范围已扩展到包括查询模型可能与最终 (后继) 模型不同的情况——这种现象被称为模型不匹配——或者根本不使用模型进行查询的情况。

但同样, 许多研究 (Baldridge and Osborne, 2004; Schröder et al., 2022; Shelmanov et al., 2021) 报告称, 查询模型与后继模型之间的不匹配常常导致不理想的结果 (Zhang et al., 2022c)。尽管 Shelmanov et al. (2021) 和 Nguyen et al. (2022) 的努力试图通过使用更小、相似的模型进行查询来解决这个问题, 但仍然存在对模型类型的依赖。类似地, Tsvigun et al. (2022) 提供了一种有前景的方法来解耦查询模型和后继模型, 尽管在这个过程中仍然需要一个蒸馏版本。

Research Gap: 与使用更小、资源高效的模型进行查询不同, 我们建议利用更大的语言模型。这种方法起初可能看起来违反直觉, 因为担心较长的运行时间, 但由于这些模型能够进行零样本学习, 因此不需要训练 (Shliazhko et al., 2024)。此外, 这些模型可以通过几个免费的聊天界面实时访问, 不需要机器学习专业知识, 使其成为实际应用的理想选择。

尽管主动学习能通过减少标注工作量显著优化学习流程, 但多数策略都面临冷启动问题。在缺乏足够初始标注数据 (标注种子) 的情况下, 主动学习系统可能难以对不确定性或多样性做出有根据的预测 (Yuan et al., 2020)。这对于低数据环境构成了重大挑战, 尤其是在可用数据实例极少的少样本场景中。Yuan et al. (2020) 提出了 ALPS, 这是为数不多直接针对基于 Transformer 模型的主动学习冷启动问题

的研究之一。ALPS 利用 BERT 已有的掩码语言建模目标并结合聚类来选择实例。作者假设, 在模型训练的初始阶段, 该策略相比分类器头部能提供更可靠的信度分数。然而, 其结果好坏参半, 这一点后来也被 Nguyen et al. (2022) 所证实。

Research Gap: 冷启动问题是主动学习研究中的一个主要挑战, 对于少样本场景尤其如此。本研究旨在通过提出一种无需初始标注数据的主动学习策略来克服这一挑战, 从而使得在主动学习流程的最初阶段就能实现性能提升。该方法也可通过为其他主动学习方法提供初始数据, 用于解决它们的冷启动问题。

2.2 Few-Shot Learning

小样本学习指的是模型需要从有限数量的训练实例中学习的方法。这些场景因其在现实世界问题中的重要性 (Yu and Raschka, 2020) 以及深度学习模型日益增强的能力而变得越来越相关。虽然像 GPT-3 这样的大型语言模型通过上下文学习天生就具备强大的小样本或零样本能力, 但我们的重点在于增强像 BERT 这样较小模型的小样本学习能力。通过特定的少样本学习方法, 如 ADAPET (Tam et al., 2021), 这些较小的语言模型可以获得与 GPT-3 相似甚至更好的小样本性能。

一种方法是将任务表述为完形填空式测试, 即文本实例中的某些词语缺失, 语言模型被用来填补这些空白。这样, 无需在语言模型之上训练分类器, 从而能更有效地利用语言模型 (Gao et al., 2021)。这种方法以 ADAPET 方法 (Tam et al., 2021) 以及 Gao et al. (2021); Zhang et al. (2022a); Schick and Schütze (2021a) 的工作为例。由 Mahabadi et al. (2022) 提出的 PERFECT 方法, 通过整合任务特定的适配器和专门的标签嵌入, 消除了手动设计此类完形填空任务的需要。类似地, 在由 Tunstall et al. (2022) 提出的 SetFit 中, 由于在可用训练数据上微调了一个句子转换器 (Reimers and Gurevych, 2019), 因此不需要完形填空任务。

Table 1: Overview of different works applying AL to transformer models. "Model" describes the successor model trained from the AL process. "Matching" indicates if the successor model matches the query model.

are then encoded with the resulting model and a regular classification head is trained on them.

While not few-shot methods per se, transfer learning and data augmentation are often employed to address few-shot scenarios. Transfer learning allows models to leverage knowledge from related domains (Ruder et al., 2019), providing a strong foundation in few-shot scenarios. Data augmentation, particularly using LLMs (Ding et al., 2024; Wang et al., 2022; Xu et al., 2024), has been used to generate high-quality synthetic data in low-data regimes (Zheng et al., 2023). Similarly, in this study we investigate whether AL can be useful for few-shot learning, especially by overcoming the cold-start problem.

Research Gap: In general, LLMs dominate the machine learning field, in particular in few-shot learning, but there are numerous reasons why researchers and practitioners still rely on smaller BERT-like models, especially for inference, where concerns such as memory limitations, API costs, and privacy often preclude the use of LLMs (Wang et al., 2024a; Samuel, 2024; Bosley et al., 2023). To the best of our knowledge, our approach is the first to leverage the capabilities of LLMs to support instance selection for training smaller BERT-like models. In doing so, we combine the flexibility and efficiency of smaller models with the strategic guidance of LLMs.

While Ein-Dor et al. (2020) and Griebhaber et al. (2020) also assess the performance of AL with BERT in low-resource scenarios, their datasets are still considerably larger than those in few-shot scenarios. However, in scenarios with very few instances, example selection has the greatest impact. In this work, we aim to improve state-of-the-art few-shot learning methods with our AL approach.

3 Method

ActiveLLM is a pool-based sampling method that operates in batch mode, meaning that it selects a subset from a pool of unlabeled data for querying an oracle. It employs instruction-tuned LLMs as query models, while allowing the choice of a successor model to be independent of these models.

A demonstration of the process is given in Figure 1, where the LLM is instructed to select instances, which are then labeled and used to train a BERT-like model, allowing any such model to be trained.

The design of the prompts is critical for achieving optimal results with instruction-tuned LLMs. To this end, we have crafted detailed prompts in various configurations to determine their effectiveness in this context. We differentiate between two modes of ActiveLLM. The first addresses few-shot scenarios, where ActiveLLM is executed only once on the dataset. The second is suitable for general scenarios involving iterative querying, which also incorporates feedback from previous iterations.

3.1 Few-Shot Learning Mode

The prompts are constructed so that the language model receives a description of AL, including exact details of its tasks, while also keeping the context length small, as this is a limiting factor for current models (Liu et al., 2024). Given a specific task, ActiveLLM creates a prompt consisting of AL role allocation, instructions on the selection process, a description of how to format the output, followed by a batch of unlabeled instances.

An excerpt of the prompt design is illustrated in Figure 2. The parameters displayed on the right indicate certain text phrase constellations which can be included, excluded, or adjusted:

Guidelines: This includes a phrase indicating that the model is given not only a set of instances but also guidelines for the task.

Instances: This parameter determines how many instances the model should label. We test different values in Section 4.2.3.

Advice: Initial experiments showed that the LLMs often respond with explaining general strategies for choosing instances in an AL manner. Hence, we included the parameter 'advice,' which incorporates these strategies.

No CoT/CoT/Explanation: One of the most common and beneficial prompt engineering strategies is the use of Chain of Thought (CoT) prompting (Wei et al., 2022). We experiment with no CoT, 'normal' CoT ("think step by step"), and the task to explain the thoughts on each instance.

实例随后用得到的模型进行编码，并在其上训练一个常规的分类头。

虽然本身并非小样本方法，但迁移学习和数据增强常被用来应对小样本场景。迁移学习允许模型利用来自相关领域的知识 (Ruder et al., 2019)，为小样本场景提供坚实的基础。数据增强，特别是使用大型语言模型 (Ding et al., 2024; Wang et al., 2022; Xu et al., 2024)，已被用于在低数据状态下生成高质量的合成数据 (Zheng et al., 2023)。类似地，在本研究中，我们探讨主动学习是否对小样本学习有用，特别是通过克服冷启动问题。

Research Gap: 总体而言，大型语言模型主导着机器学习领域，尤其是在小样本学习中，但研究人员和实践者仍有诸多理由依赖较小的类 BERT 模型，特别是在推理阶段，因为内存限制、API 成本和隐私等问题常常排除了使用大型语言模型的可能性 (Wang et al., 2024a; Samuel, 2024; Bosley et al., 2023)。据我们所知，我们的方法是首个利用大型语言模型的能力来支持为训练较小的类 BERT 模型进行实例选择的方法。通过这样做，我们将较小模型的灵活性和效率与大型语言模型的战略指导相结合。然而，在实例数量极少的情况下，示例选择的影响最为显著。在本工作中，我们旨在通过我们的主动学习方法改进最先进的少样本学习方法。

3 Method

ActiveLLM 是一种基于池的批量模式采样方法，这意味着它从一组未标记数据池选择一个子集来向标注者 (oracle) 进行查询。它采用指令微调的大型语言模型作为查询模型，同时允许后续模型的选择独立于这些模型。该过程的演示如图1所示，其中指示大型语言模型选择实例，随后对这些实例进行标注并用于训练一个类似 BERT 的模型，从而使得任何此类模型都能被训练。

提示的设计对于利用指令微调的大型语言模型

获得最佳结果至关重要。为此，我们精心设计了多种配置の詳細提示，以确定它们在此背景下的有效性。我们区分了 ActiveLLM 的两种模式。第一种针对少样本场景，其中 ActiveLLM 仅在数据集上执行一次。第二种适用于涉及迭代查询的一般场景，该模式还整合了先前迭代的反馈。

3.1 Few-Shot Learning Mode

提示的构建方式使得语言模型能够接收到关于主动学习 (AL) 的描述，包括其任务的确切细节，同时保持较短的上下文长度，因为这是当前模型的一个限制因素 (Liu et al., 2024)。给定一个特定任务，ActiveLLM 会创建一个提示，该提示包含 AL 角色分配、选择过程的说明、输出格式的描述，随后是一批未标记的实例。

提示设计的摘录如图 2 所示。右侧显示的参数表示某些可以包含、排除或调整的文本短语组合：

Guidelines: 这包括一个短语，表明模型不仅获得一组实例，还获得了任务指南。

Instances: 此参数决定模型应标记多少个实例。我们在第4.2.3节测试了不同的值。

Advice: 初步实验表明，大语言模型 (LLMs) 经常以解释以 AL 方式选择实例的一般策略来回应。因此，我们加入了 'advice' 参数，该参数包含了这些策略。

No CoT/CoT/Explanation: 最常见且有益的提示工程策略之一是使用思维链 (CoT) 提示 (Wei et al., 2022)。我们尝试了无 CoT、'普通' CoT ("逐步思考") 以及解释每个实例思考过程的任务。

最后，一批未标记的实例被附加到提示中。我们假设，与常见的 AL 策略相反，为了找到具有高学习影响力的实例，不需要查询模型评估大批量甚至所有实例。我们在第4.2.2节测试了此参数的不同值。

Consider yourself in the position of an active learning component to help a human annotator. You have to choose the instances that the annotator has to label. You are given (the guidelines for the task and) a set of instances of a dataset. You can only choose (32) instances. (You would ideally want to choose those instances that would provide the most informative and diverse data for the model. Here are some strategies to consider: *advice*)

(Label Guidelines for the human annotator: *guidelines*)

(The output format should be a list of the instances that you would label, separated by a comma.) (Please think step by step about what you would do to select the instances to label. After this provide the list of instances that you would label, separated by a comma.) (Please describe instance by instance why you would select or not select it for labelling. Do not stop before you successfully found (32) that you would suggest to label. After this provide the list of the instances that you would label, separated by a comma.)

For example, if you would label the instances 1, 4, 5 then the output should be: 1, 4, 5. The following instances are given to you (separated with "\n ##### \n"):

Guidelines
Instances
Advice

Guidelines

No CoT
Normal CoT

Explanation
Instances

Figure 2: Prompt design of ActiveLLM in few-shot learning mode.

Consider yourself in the position of an active learning component to help a human annotator. You have to choose the instances that the annotator has to label. You are given (the guidelines for the task and) a set of instances of a dataset. You can only choose (32) instances. (You would ideally want to choose those instances that would provide the most informative and diverse data for the model. Here are some strategies to consider: *advice*)

(Label Guidelines for the human annotator: *guidelines*)

(The output format should be a list of the instances that you would label, separated by a comma.) (Please think step by step about what you would do to select the instances to label. After this provide the list of instances that you would label, separated by a comma.) (Please describe instance by instance why you would select or not select it for labelling. Do not stop before you successfully found (32) that you would suggest to label. After this provide the list of the instances that you would label, separated by a comma.)

For example, if you would label the instances 1, 4, 5 then the output should be: 1, 4, 5. The following instances are given to you (separated with "\n ##### \n"):

Guidelines
Instances
Advice

Guidelines

No CoT
Normal CoT

Explanation
Instances

图 2: ActiveLLM 在少样本学习模式下的提示设计。

Finally, a batch of unlabeled instances is appended to the prompt. We hypothesize that, contrary to common AL strategies, large batches or even all instances do not need to be evaluated by the query model to find instances with high learning impact. We test different values of this parameter in Section 4.2.2.

3.2 Iterated Querying Mode

In the second variant, we design ActiveLLM similar to other AL strategies, allowing the LLM to query new instances repeatedly. For this, it is sensible to include data that was selected in previous iterations. As the prompt should not become too large, we consider three additional parameters (see Appendix A and Figure 9 for the full prompt):

No Recap: Indicating the baseline, where no instances from previous queries are recalled.

Recap: The instances from previous queries are directly included in the prompt.

Index Recap: To reduce the context size, the indices of the instances from previous queries are included in the prompt.

For the latter two scenarios, we included instructions in the prompt to indicate that instances from previous iterations are incorporated. In both cases, we did not include the annotated labels, as our experiments showed that they were generally disregarded by the language models. An example of a specific prompt and LLM answer is given in Appendix Figure 10.

4 Experiments

4.1 Description

Our experimentation perspective with ActiveLLM focuses on addressing the following questions:

1. How should the prompt be designed? (Section 4.2)
2. Can the chosen prompt be applied to other models and datasets? (Section 4.3)
3. How does ActiveLLM compare to other AL strategies? (Section 4.4)
4. Can ActiveLLM improve state-of-the-art few-shot learning methods? (Section 4.5)
5. How does the method perform in non-few-shot-scenarios? (Section 4.6.1)
6. Is ActiveLLM capable of overcoming the cold-start problem in other AL strategies? (Section 4.6.2)

As is common in research, our experiments simulate the AL cycle. To do this, we take a subset of the training set (labels removed) for the respective task and allow the LLM to select a certain number of instances based on the prompt given in Section 3. These instances are then assigned the true labels as if they had been annotated by a perfect annotator.

4.1.1 Datasets

For the initial prompt engineering experiments, we are interested in a less common dataset, as an LLM might be less biased by data leakage. We chose the Specialized CTI dataset (Bayer et al., 2023), where Twitter posts are classified according to their relevance to specific CTI events (further details in Appendix B). In the general applicability tests, we used the best-performing prompt from the prompt engineering phase on the commonly-used GLUE benchmark. For comparison with the

3.2 Iterated Querying Mode

在第二种变体中，我们将 ActiveLLM 设计为类似于其他主动学习策略，允许大语言模型反复查询新实例。为此，纳入先前迭代中选择的实例是合理的。由于提示不应变得过于庞大，我们考虑了三个附加参数（完整提示参见附录A与图9）：

No Recap: 表示基线情况，即不回忆先前查询中的任何实例。

Recap: 先前查询中的实例直接包含在提示中。

Index Recap: 为减少上下文长度，将先前查询中实例的索引包含在提示中。

针对后两种场景，我们在提示中加入了说明，以表明先前迭代的实例已被纳入。在这两种情况下，我们均未包含已标注的标签，因为实验表明语言模型通常忽略这些标签。具体提示及大语言模型回答的示例见附录图10。

4 Experiments

4.1 Description

我们针对 ActiveLLM 的实验视角聚焦于解决以下问题：

1. 应如何设计提示？（章节 4.2）
2. 所选提示能否应用于其他模型和数据集？（章节 4.3）
3. ActiveLLM 与其他主动学习策略相比如何？（章节 4.4）

4. ActiveLLM 能否改进最先进的少样本学习方法？（章节 4.5）

5. 该方法在非少样本场景下表现如何？（章节 4.6.1）

6. ActiveLLM 能否克服其他主动学习策略中的冷启动问题？（章节 4.6.2）

正如研究中常见的那样，我们的实验模拟了主动学习循环。为此，我们取相应任务训练集的一个子集（标签已移除），并允许 LLM 根据章节 3 中给出的提示选择一定数量的实例。然后，这些实例被分配真实标签，如同它们已由完美标注者标注完成。

4.1.1 Datasets

在初始的提示工程实验中，我们关注一个较少见的数据集，因为 LLM 可能较少受到数据泄露带来的偏见影响。我们选择了 Specialized CTI 数据集 (Bayer et al., 2023)，其中推文根据其特定 CTI 事件的相关性进行分类（更多细节见附录 B）。在通用性测试中，我们将提示工程阶段表现最佳的提示应用于常用的 GLUE 基准测试。为了与少样本学习方法 SetFit 进行比较，我们使用 AGNews 数据集 (Zhang et al., 2015)，因为该方法有适用于此数据集的适配版本。SST-2 数据集 (Socher et al., 2013) 先前已在通用性部分使用，在非少样本实验中也同样被采用。为衡量性能，我们使用 CTI 和 AGNews 的测试集，以及 GLUE 的验

few-shot learning method SetFit, we use the AG-News dataset (Zhang et al., 2015), for which an adaptation of the method is available. The SST-2 dataset (Socher et al., 2013), previously used in the general applicability section, is also employed in the non-few-shot experiments. To measure the performance we use the test sets for CTI and AG-News, and the validation sets for GLUE. Unless otherwise specified (see Sections 4.2.3 and 4.6), we select 32 instances per task, either via AL or random sampling for the baseline, which represents a common few-shot learning setting (Tam et al., 2021; Schick and Schütze, 2021b; Mahabadi et al., 2022).

During the prompt engineering experiments, we aim to minimize random factors by generating five different randomizations of the CTI task and applying both the baseline and each prompt to these variations. For each evaluation, we select five different seeds for training the successor model, with the reported results for the baseline and each prompt being the average score from 25 runs (5 dataset randomizations times 5 trained models). For subsequent experiments, we create one randomization of the datasets but continue to train five models and provide the average score.

4.1.2 Models and Hyperparameters

In our experiments, we primarily employed GPT-4 (OpenAI, 2023) as the query model for ActiveLLM, as it achieved top results at the time the main experiments were conducted. However, in Section 4.3 we also experiment with other LLMs: o1 (OpenAI, 2024), GPT-4o, GPT-3.5 (Brown et al., 2020), Llama 3 70B, Gemini-Ultra 1.0 (Anil et al., 2023), Mistral Large, and Mixtral 8x7B (Jiang et al., 2024) (ActiveLLM is renamed according to the used LLM - e.g. GPT-4: ActiveGPT4). To underscore the practicality of our method, we utilize chat versions of the LLMs to simulate realistic interactions within chat environments (further details in Appendix C).

As a successor model for the selected instances of ActiveLLM, we chose BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019). While foundational works such as from Devlin et al. (2019) typically train BERT-like models for only 3-5 epochs, other studies (Gao et al., 2021; Mosbach et al., 2021) suggest that higher epoch counts are necessary for stable and optimal performance in low-data regimes. Consequently, we have opted for 25 epochs for our experiments. Otherwise, we

adhere to the default Huggingface transformer parameters, with a learning rate of $1e-3$, a weight decay of 0.01, and a dropout probability of 0.1.

For comparison, we include in each experiment a random sampling baseline where the first instances from a shuffled dataset are selected. In addition, we compare ActiveLLM with the most commonly used AL strategies for transformers identified in our literature overview (see Table 2.1.3): LC, BALD, EKM, and PE. The strategies are implemented in a model-matching scenario (i.e., training the same query and successor model), can query the entire dataset of each task, and select 5 instances until the selection of 32 is reached. ActiveLLM, on the other hand, is implemented in a model-mismatch scenario, can only query a subset of the full training set, and selects the 32 instances in one query (unless otherwise specified). To implement the common AL strategies, we use the small-text library (Schröder et al., 2023), which provides a unified framework for pool-based AL with support for transformer-based models.

4.2 Prompt Engineering Experiments

- How should the prompt be designed?

To answer the first question, we experiment with different configurations of ActiveLLM, such as whether to include AL strategy advices or guidelines, as shown in Figure 2. We then evaluate the effects of varying batch sizes for the unlabeled data used in the prompts. Finally, we compare different selection sizes, i.e. the number of instances the LLM should select. While we want to find the most appropriate prompt, our goal is not to engineer every detail of it, which is left open for future work and practitioners.

4.2.1 Configuration Results

Table 2 presents the results of various prompt configurations. For these, we use a batch of 300 presented unlabeled instances. Prompt configurations A and B differ in whether the advice on how to select the instances is already included in the prompt. We observed that GPT-4 tends to reiterate the advice even when it is provided in the prompt. Furthermore, the results suggest that it is preferable not to include the advice in the prompt, likely due to the increased context size when the advice is repeated in both the prompt and the response.

We also test the three CoT variants: A1 and B1, where there is no CoT; A2 and B2, which use

证集。除非另有说明（见章节 4.2.3 和 4.6），我们为每个任务选择 32 个实例，通过主动学习或基线随机抽样选取，这代表了一种常见的少样本学习设置 (Tam et al., 2021; Schick and Schütze, 2021b; Mahabadi et al., 2022)。

在提示工程实验期间，我们旨在通过生成 CTI 任务的五种不同随机化版本，并将基线及每种提示应用于这些变体，以最小化随机因素的影响。对于每次评估，我们为训练后续模型选择五种不同的随机种子，所报告的基线和每种提示的结果均为 25 次运行（5 种数据集随机化 5 个训练模型）的平均得分。在后续实验中，我们创建数据集的一种随机化版本，但继续训练五个模型并提供平均得分。

4.1.2 Models and Hyperparameters

在我们的实验中，我们主要采用 GPT-4 (OpenAI, 2023) 作为 ActiveLLM 的查询模型，因为它在主要实验进行时取得了最佳结果。然而，在章节 4.3 中，我们也尝试了其他大语言模型：o1 (OpenAI, 2024)、GPT-4o、GPT-3.5 (Brown et al., 2020)、Llama 3 70B、Gemini-Ultra 1.0 (Anil et al., 2023)、Mistral Large 和 Mixtral 8x7B (Jiang et al., 2024) (ActiveLLM 会根据使用的大语言模型重新命名——例如 GPT-4: ActiveGPT4)。为了强调我们方法的实用性，我们使用大语言模型的聊天版本来模拟聊天环境中的真实交互（更多细节见附录 C）。

作为 ActiveLLM 所选实例的后继模型，我们选择了 BERT (Devlin et al., 2019) 和 RoBERTa (Liu et al., 2019)。虽然如 Devlin et al. (2019) 等基础性工作通常只训练 BERT 类模型 3-5 个轮次，但其他研究 (Gao et al., 2021; Mosbach et al., 2021) 表明，在低数据量情况下，需要更高的轮次数才能获得稳定且最优的性能。因此，我们为实验选择了 25 个轮次。除此之外，我们遵循 Huggingface transformer 的默认参数，学习率为 $1e-3$ ，权重衰减为 0.01，丢弃概率为 0.1。

为了进行比较，我们在每个实验中包含了一个随机采样基线，即从打乱的数据集中选择前几个实例。此外，我们将 ActiveLLM 与我们文献综述中确定的、针对 transformer 最常用的主动学习策略（见表 ??）进行比较：LC、BALD、EKM 和 PE。这些策略在模型匹配场景中实现（即训练相同的查询模型和后继模型），可以查询每个任务的整个数据集，并选择 5 个实例，直到达到 32 个的选择量。另一方面，ActiveLLM 在模型不匹配场景中实现，只能查询完整训练集的一个子集，并且通过一次查询选择 32 个实例（除非另有说明）。为了实现常见的主动学习策略，我们使用了 small-text 库 (Schröder et al., 2023)，它提供了一个统一的框架，支持基于 transformer 的模型进行基于池的主动学习。

4.2 Prompt Engineering Experiments

- How should the prompt be designed?

为了回答第一个问题，我们尝试了 ActiveLLM 的不同配置，例如是否包含主动学习策略建议或指导原则，如图 2 所示。然后，我们评估了提示中使用的未标记数据的不同批次大小的影响。最后，我们比较了不同的选择规模，即大语言模型应选择的实例数量。虽然我们希望找到最合适的提示，但我们的目标并非精心设计其每一个细节，这部分留待未来工作和实践者探索。

4.2.1 Configuration Results

表 1 展示了各种提示配置的结果。对于这些配置，我们使用了一批 300 个未标记的实例。提示配置 A 和 B 的区别在于关于如何选择实例的建议是否已包含在提示中。我们观察到，即使提示中提供了建议，GPT-4 也倾向于重申该建议。此外，结果表明最好不要在提示中包含建议，这可能是由于建议在提示和响应中重复出现时增加了上下文长度。

我们还测试了三种思维链变体：A1 和 B1，不使用思维链；A2 和 B2，使用标准的思维链方

	A	CoT	G	F1 (SD)
Baseline	-	-	-	0.6781 (0.029)
Prompt A1	✓	✗	✗	0.6522 (0.100)
Prompt A2	✓	✓	✗	0.6951 (0.102)
Prompt A3	✓	✓+	✗	0.6876 (0.108)
Prompt B1	✗	✗	✗	0.7135 (0.060)
Prompt B2	✗	✓	✗	0.7214 (0.053)
Prompt B3	✗	✓+	✗	0.6271 (0.139)
Prompt C2	✗	✓	✓	0.7329 (0.053)

Table 2: Prompt engineering results of the baseline and ActiveGPT4 on the CTI dataset sampling 32 instances. The results are averaged over 25 runs (5 random dataset and 5 random model initializations). A (advice), G (guidelines), C (no advice, but guidelines), 1 (no CoT), 2 (CoT: step-by-step), 3 (CoT+: explain each instance).

the standard CoT approach ("think step by step"); and A3 and B3, which instruct to include explanations for each instance. The common "think step by step" instruction yielded the best results. This might be because this approach allows GPT more reasoning in structured steps, while not increasing the context size too much.

Finally, we experimented with including labeling guidelines in the prompt. As anticipated, this prompt configuration produced the best results, likely because it enabled the model to better differentiate between instances. Although labeling guidelines are often created prior to the labeling process, they are rarely included in the dataset for common tasks. While the guidelines are available for the CTI task, its inclusion in the prompt should be seen as an exploration of what can be achieved rather than the default case for ActiveLLM. Therefore, in future evaluations, we will utilize ActiveGPT4 with prompt B2, i.e., no guidelines or additional advice on AL, only employing the simple "think step by step" CoT prompting.

4.2.2 Presented Batch Size Results

In the subsequent experiment, we explore varying batch sizes of unlabeled instances presented to ActiveGPT4. Figure 3 displays the results for 50, 100, 200, 300, and 400 examples. There is a noticeable trade-off between the context length and the number of presented examples. While a larger pool of examples generally allows for a more diverse set, an increase in context length tends to degrade the model's performance. It can be observed that the optimal prompt with 300 examples, as used in the previous section, could be signifi-

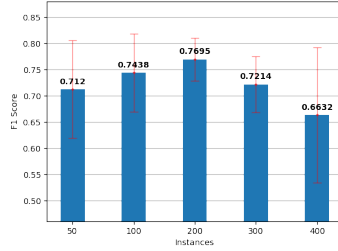


Figure 3: Varying size of the batch of unlabeled instances from which ActiveGPT4 can select 32 instances. All results are averaged over 5 runs on the CTI dataset (F1).

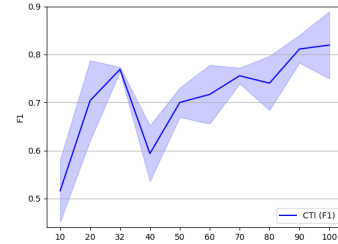


Figure 4: Varying selection size of examples to be selected by ActiveGPT4. All results (F1) are averaged over 5 runs on the CTI dataset.

cantly enhanced by reducing the size to 200. Consequently, for the primary experiments in Section 4.3, we limited the number of instances presented to the model to 200, contingent on the model's capacity to handle such a context size.

4.2.3 Selection Size Results

In our final experiment on prompt engineering, we investigate the impact of the selection size—how many instances the model should choose—on the outcomes. Many research works (Tam et al., 2021; Schick and Schütze, 2021b; Mahabadi et al., 2022) typically use 32 examples for few-shot scenarios. We evaluate whether smaller or larger sizes might be more effective for ActiveGPT4. Figure 4 illustrates varying selection sizes for the CTI dataset. Selecting 32 examples per query appears to be a sensible choice. Notably, performance declines

	A	CoT	G	F1 (SD)
Baseline	-	-	-	0.6781 (0.029)
Prompt A1	✓	✗	✗	0.6522 (0.100)
Prompt A2	✓	✓	✗	0.6951 (0.102)
Prompt A3	✓	✓+	✗	0.6876 (0.108)
Prompt B1	✗	✗	✗	0.7135 (0.060)
Prompt B2	✗	✓	✗	0.7214 (0.053)
Prompt B3	✗	✓+	✗	0.6271 (0.139)
Prompt C2	✗	✓	✓	0.7329 (0.053)

表 1: 基线和 ActiveGPT4 在 CTI 数据集上提示工程的结果 (采样 32 个实例)。结果是 25 次运行 (5 次随机数据集和 5 次随机模型初始化) 的平均值。A (建议), G (指导原则), C (无建议, 但有指导原则), 1 (无思维链), 2 (思维链: 逐步推理), 3 (思维链+: 解释每个实例)。

法 ("逐步思考"); 以及 A3 和 B3, 要求包含对每个实例的解释。常见的 "逐步思考" 指令产生了最佳结果。这可能是因为这种方法允许 GPT 以结构化的步骤进行更多推理, 同时不会过多增加上下文长度。

最后, 我们尝试在提示中包含标注指导原则。正如预期, 这种提示配置产生了最佳结果, 可能是因为它使模型能够更好地区分实例。尽管标注指导原则通常在标注过程之前制定, 但在常见任务的数据集中很少包含它们。虽然 CTI 任务提供了指导原则, 但将其包含在提示中应被视为探索可能达到的效果, 而非 ActiveLLM 的默认情况。因此, 在未来的评估中, 我们将使用 ActiveGPT4 并采用提示 B2, 即不包含指导原则或关于主动学习的额外建议, 仅使用简单的 "逐步思考" 思维链提示。

4.2.2 Presented Batch Size Results

在后续实验中, 我们探索了呈现给 ActiveGPT4 的未标记实例批次大小的变化。图3展示了 50、100、200、300 和 400 个示例的结果。可以观察到上下文长度与呈现示例数量之间存在明显的权衡。虽然更大的示例池通常允许更丰富的多样性, 但上下文长度的增加往往会降低模型的性能。可以看出, 前一节中使用的包含 300 个示例的最优提示, 通过将数量减少到 200 个可以得到显著提升。因此, 在

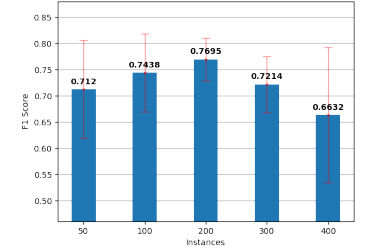


图 3: ActiveGPT4 可选择 32 个实例的未标记实例批次大小变化。所有结果均为 CTI 数据集上 5 次运行的平均值 (F1 分数)。

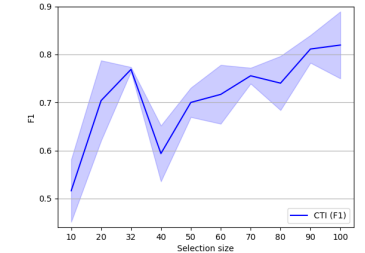


图 4: ActiveGPT4 选择示例的选取数量变化。所有结果 (F1 分数) 均为 CTI 数据集上 5 次运行的平均值。

第4.3节的主要实验中, 我们将呈现给模型的实例数量限制在 200 个, 这取决于模型处理此类上下文大小的能力。

4.2.3 Selection Size Results

在关于提示工程的最后一项实验中, 我们研究了选取数量——即模型应选择多少个实例——对结果的影响。许多研究工作 (Tam et al., 2021; Schick and Schütze, 2021b; Mahabadi et al., 2022) 通常在少样本场景中使用了 32 个示例。我们评估了更小或更大的选取数量是否可能对 ActiveGPT4 更有效。图4展示了 CTI 数据集上不同选取数量的情况。每次查询选择 32 个示例似乎是一个合理的选择。值得注意的是, 当示例数量增加到 90 个时, 性能会下

with up to 90 examples, suggesting that the model may struggle to process such a large number of examples effectively. However, performance increases beyond this point, as expected, because a larger training set is inherently beneficial.

As part of the following experiments, we also tested the selection size on the SST-2 dataset. The results are equivalent and can be found in Appendix Figure 8.

4.3 General Applicability

- Can the chosen prompt be applied to other models and datasets?

In this section, we aim to assess how ActiveLLM equipped with various LLMs and the prompt configuration B2 from Section 4.2.1 performs on the more common datasets of the GLUE benchmark.

The results of this experiment are given in Table 3. ActiveGPT4, ActiveGPT4o, Activeo1 and ActiveMistralLarge are given 200 unlabeled instances as identified in the batch size experiments (Section 4.2.2). Llama 3, GPT-3.5, Gemini-Ultra and Mistral can not process such extensive prompts and are therefore given 100 instances. All LLMs show that they are very good query models for ActiveLLM, outperforming the baseline in most cases. However, we recommend using ActiveLLM with GPT-4 or GPT-3.5, as they consistently improve results across all tasks. An interesting finding is that Mistral Large with 200 examples performed worse than its 100 example counterpart in five out of eight tasks. The tasks where Mistral Large performs better with 200 instances are those with generally shorter text length, suggesting that choosing a batch size based on token length may be better than a static batch size. This would also explain the variation in improvements across tasks and models, as some tasks involve many more tokens per instance, and some models perform better with larger prompts. In addition, the differences between the models may be due to the fact that the prompt design phase was carried out using GPT-4. Other models may work better and give more consistent results with a prompt design specifically for them. The most significant improvements can be observed with ActiveMistralLarge (200) and ActiveGPT4 on WNLI and SST-2, respectively, achieving improvements of 17.19 and 17.11 percentage points over the baseline.

Overall, ActiveLLM, utilizing GPT-4 and GPT-3.5, proves to be a robust approach for enhancing performance in few-shot scenarios across all tasks. While it may even produce better results with other LLMs, the improvements may be inconsistent and should be compared to a baseline.

To verify that our results are not specific to BERT, we repeated the experiments from Table 3 using RoBERTa. The results, shown in Appendix Table 7, confirm that ActiveLLM also improves performance on a different BERT-based architecture. As with BERT, GPT-4 and GPT-3.5 give more consistent results than LLaMA, Mistral/Mixtral, and Gemini-Ultra.

4.4 Comparison

- How does ActiveLLM compare to other AL strategies?

We are interested to see how ActiveLLM performs against other AL methods. In Table 3 we also included the results for the most used AL strategies with transformers identified in our literature overview (see Table 2.1.3): LC, BALD, EKM, and PE. While these methods are implemented with the best preconditions with a model-matching scenario and access to the whole dataset of each task, they are often not able to improve over the baseline and in every case worse than an LLM-variant of ActiveLLM. This outcome is expected, as these AL strategies face a cold-start problem. Therefore, in Section 4.6.2 we test whether ActiveLLM can be used to overcome the cold-start problem of these techniques. Furthermore, the practical use of these methods is limited, as the selection of examples to be labeled required at least several minutes and, in the case of larger datasets, several hours (see Table 4). For instance, with BALD on the QQP dataset, the approach did not yield any result even after 48 hours of execution. In contrast, ActiveLLM eliminates the need for model training during the AL process and only considers a small subset of the unlabeled data, enabling query times of just a few seconds while still achieving higher improvements.

Standard deviations for the results of Table 3 are given in the Appendix Table 6. While the values vary across tasks and models, the overall trend indicates that ActiveLLM does not amplify performance instability, and in many cases, can reduce it substantially.

降, 这表明模型可能难以有效处理如此大量的示例。然而, 超过这个数量后, 性能会如预期般提升, 因为更大的训练集本身是有益的。作为后续实验的一部分, 我们还在 SST-2 数据集上测试了选取数量。结果类似, 可在附录图8中找到。

4.3 General Applicability

- Can the chosen prompt be applied to other models and datasets?

在本节中, 我们旨在评估配备不同 LLM 的 ActiveLLM 以及第4.2.1节中的提示配置 B2 在 GLUE 基准测试更常见数据集上的表现。本实验的结果如表2所示。根据批量大小实验(第4.2.2节)中确定的结果, ActiveGPT4、ActiveGPT4o、Activeo1 和 ActiveMistralLarge

被给予 200 个未标记实例。Llama 3、GPT-3.5、Gemini-Ultra 和 Mistral 无法处理如此大量的提示, 因此被给予 100 个实例。所有 LLM 都表明它们是 ActiveLLM 的优秀查询模型, 在大多数情况下优于基线。然而, 我们建议使用 GPT-4 或 GPT-3.5 配合 ActiveLLM, 因为它们在所有任务中都能持续提升结果。一个有趣的发现是, 在八个任务中的五个任务中, 使用 200 个示例的 Mistral Large 表现不如使用 100 个示例的版本。Mistral Large 在使用 200 个实例时表现更好的任务是那些通常文本长度较短的任务, 这表明基于令牌长度选择批量大小可能比静态批量大小更好。这也解释了不同任务和模型之间改进程度的差异, 因为某些任务每个实例涉及更多令牌, 而某些模型在处理更大提示时表现更好。此外, 模型之间的差异可能是由于提示设计阶段是使用 GPT-4 进行的。其他模型如果使用专门为其设计的提示, 可能会工作得更好并给出更一致的结果。最显著的改进体现在 ActiveMistralLarge (200) 和 ActiveGPT4 上, 分别在 WNLI 和 SST-2 任务中取得了比基线高出 17.19 和 17.11 个百分点的改进。

总体而言, 利用 GPT-4 和 GPT-3.5 的

ActiveLLM 被证明是一种在所有任务的少样本场景中提升性能的稳健方法。虽然使用其他 LLM 甚至可能产生更好的结果, 但改进可能不一致, 应与基线进行比较。

为了验证我们的结果并非特定于 BERT, 我们使用 RoBERTa 重复了表2中的实验。结果如附录表6所示, 证实 ActiveLLM 也能在不同的基于 BERT 的架构上提升性能。与 BERT 的情况类似, GPT-4 和 GPT-3.5 比 LLaMA、Mistral/Mixtral 和 Gemini-Ultra 给出更一致的结果。

4.4 Comparison

- How does ActiveLLM compare to other AL strategies?

我们关注 ActiveLLM 与其他主动学习方法(AL)的性能对比。在表2中, 我们还纳入了文献综述(见表??)中识别出的、与 Transformer 结合最常用的几种 AL 策略的结果: LC、BALD、EKM 和 PE。尽管这些方法在模型匹配场景和可访问各任务完整数据集的最佳前提下实施, 但它们通常无法超越基线, 并且在所有情况下都逊于 ActiveLLM 的 LLM 变体。这一结果是预期的, 因为这些 AL 策略面临着冷启动问题。因此, 在章节4.6.2中, 我们测试了 ActiveLLM 是否可用于克服这些技术的冷启动问题。此外, 这些方法的实际应用有限, 因为选择待标注样本至少需要数分钟, 对于较大的数据集则需要数小时(见表3)。例如, 在 QQP 数据集上使用 BALD 方法, 即使执行 48 小时后也未产生任何结果。相比之下, ActiveLLM 消除了 AL 过程中模型训练的需求, 并且仅考虑未标注数据的一个小子集, 从而实现了仅需数秒的查询时间, 同时仍能获得更高的性能提升。

表2结果的标准差见附录表5。虽然数值在不同任务和模型间有所变化, 但总体趋势表明 ActiveLLM 不会放大性能不稳定性, 并且在许多情况下能显著降低不稳定性。

		QNLI	QQP	RTE	SST2	WNLI	MNLI-(m/mm)	MRPC	COLA
	Baseline	55.11	63.67	50.83	56.03	35.77	34.92/35.01	67.55	63.89
200	GPT-4	58.94 ↑	63.99 ↑	53.57 ↑	73.14 ↑	40.28 ↑	35.45/35.14 ↑	68.33 ↑	64.70 ↑
	GPT-4o	62.76 ↑	63.65 ↓	48.88 ↓	66.31 ↑	41.13 ↑	32.23/32.80 ↓	69.75 ↑	65.41 ↑
	o1	62.10 ↑	64.00 ↑	54.01 ↑	59.08 ↑	39.43 ↑	33.52/33.94 ↓	63.87 ↓	69.91 ↑
	Mistral Large	60.23 ↑	65.26 ↑	55.60 ↑	60.64 ↑	52.96 ↑	36.59/35.92 ↑	67.06 ↓	67.44 ↑
100	Llama 3 70B	56.57 ↑	65.82 ↑	52.27 ↑	65.71 ↑	40.28 ↑	37.19/36.68 ↑	66.86 ↓	68.15 ↑
	Mistral Large	60.68 ↑	64.82 ↑	49.48 ↓	66.86 ↑	46.48 ↑	37.41/36.58 ↑	69.07 ↑	67.56 ↑
	GPT-3.5	55.11 -	65.88 ↑	55.09 ↑	61.15 ↑	51.55 ↑	37.44/37.62 ↑	67.70 ↑	68.63 ↑
	Gemini-Ultra	57.60 ↑	61.18 ↓	54.58 ↑	49.08 ↓	43.10 ↑	35.74/35.77 ↑	69.95 ↑	69.22 ↑
AL	Mixtral 8x7B	61.13 ↑	65.27 ↑	50.97 ↑	68.00 ↑	41.41 ↑	33.62/33.75 ↓	68.14 ↑	66.96 ↑
	LC	56.76 ↑	58.65 ↓	54.44 ↑	56.86 ↑	42.82 ↑	36.48/36.18 ↑	66.96 ↓	69.26 ↑
	BALD	59.91 ↑	n.a.*	52.13 ↑	69.17 ↑	44.79 ↑	36.14/36.34 ↑	68.53 ↑	64.24 ↑
	EKM	52.44 ↓	59.62 ↓	47.65 ↓	69.77 ↑	46.49 ↑	32.87/32.75 ↓	61.08 ↓	66.60 ↑
	PE	53.75 ↓	58.65 ↓	54.44 ↑	56.86 ↑	42.82 ↑	37.11/37.20 ↑	66.96 ↓	69.26 ↑

Table 3: BERT-base few-shot results (32 instances) of ActiveLLM with different LLMs and of other AL strategies (LC, BALD, EKM, and PE) on GLUE tasks. ‘Active 200/100’ describes the size of the batch of unlabeled instances. All results are averaged over 5 runs (accuracy). Standard deviations are given in Appendix Table 6.

*No result available after 48 hours of execution.

		QNLI	QQP	RTE	SST2	WNLI	MNLI	MRPC	COLA
	Average	265.26	279.11	11.01	170.34	5.17	627.12	12.31	25.29
AL	LC	81.58	271.40	6.39	53.11	3.55	297.00	5.50	11.07
	BALD	812.26	n.a.*	22.34	520.11	8.37	1610.23	30.54	68.26
	EKM	85.16	289.39	7.23	53.53	4.40	306.43	6.08	10.02
	PE	82.02	276.54	6.47	53.01	4.37	293.24	5.58	10.22

Table 4: Runtime of the AL strategies (LC, BALD, EKM, and PE) on the GLUE tasks from Table 3. The runtime (in minutes) refers to the total time required to select 32 instances over 6 AL iterations.

*No result available after 48 hours of execution.

4.5 ActiveLLM and Few-Shot Learning

- Can ActiveLLM improve state-of-the-art few-shot learning methods?

While showing significant improvements in few-shot scenarios compared to random sampling and other AL strategies, one might ask why not directly use few-shot learning methods. Therefore, we evaluate the performance of ActiveGPT4 both individually and in combination with the state-of-the-art few-shot learning methods ADAPET, PERFECT, and SetFit. For each method, we have to use a different dataset for which an implementation is available: ADAPET with CTI, PERFECT with SST-2¹, and SetFit with AgNews.

The results of this experiment are shown in Figure 5. ActiveGPT4 significantly outperforms the few-shot learning methods ADAPET and PERFECT (+ 17.64% + 24.29%). While not out-

¹The original implementation only supports RoBERTa-Large.

performing SetFit on its own, ActiveLLM, when combined with each few-shot learning method, significantly improves all of these methods. This shows that few-shot learning methods can be improved through the selection of instances, and in some cases, this selection is even more important than the use of few-shot learning methods.

4.6 Non-Few-Shot-Scenarios

In our last experiments, we investigate how ActiveLLM performs in non-few-shot scenarios and if it can be used to overcome the cold-start problem of other AL strategies. For this purpose, we select a sample size of 25 examples per AL iteration and evaluate the process up to 300 examples.

4.6.1 Iterated Querying

- How does the method perform in non-few-shot scenarios?

Figure 6 indicates that ActiveGPT4 is particularly beneficial in few-shot scenarios up to about

		QNLI	QQP	RTE	SST2	WNLI	MNLI-(m/mm)	MRPC	COLA
	Baseline	55.11	63.67	50.83	56.03	35.77	34.92/35.01	67.55	63.89
200	GPT-4	58.94 ↑	63.99 ↑	53.57 ↑	73.14 ↑	40.28 ↑	35.45/35.14 ↑	68.33 ↑	64.70 ↑
	GPT-4o	62.76 ↑	63.65 ↓	48.88 ↓	66.31 ↑	41.13 ↑	32.23/32.80 ↓	69.75 ↑	65.41 ↑
	o1	62.10 ↑	64.00 ↑	54.01 ↑	59.08 ↑	39.43 ↑	33.52/33.94 ↓	63.87 ↓	69.91 ↑
	Mistral Large	60.23 ↑	65.26 ↑	55.60 ↑	60.64 ↑	52.96 ↑	36.59/35.92 ↑	67.06 ↓	67.44 ↑
100	Llama 3 70B	56.57 ↑	65.82 ↑	52.27 ↑	65.71 ↑	40.28 ↑	37.19/36.68 ↑	66.86 ↓	68.15 ↑
	Mistral Large	60.68 ↑	64.82 ↑	49.48 ↓	66.86 ↑	46.48 ↑	37.41/36.58 ↑	69.07 ↑	67.56 ↑
	GPT-3.5	55.11 -	65.88 ↑	55.09 ↑	61.15 ↑	51.55 ↑	37.44/37.62 ↑	67.70 ↑	68.63 ↑
	Gemini-Ultra	57.60 ↑	61.18 ↓	54.58 ↑	49.08 ↓	43.10 ↑	35.74/35.77 ↑	69.95 ↑	69.22 ↑
AL	Mixtral 8x7B	61.13 ↑	65.27 ↑	50.97 ↑	68.00 ↑	41.41 ↑	33.62/33.75 ↓	68.14 ↑	66.96 ↑
	LC	56.76 ↑	58.65 ↓	54.44 ↑	56.86 ↑	42.82 ↑	36.48/36.18 ↑	66.96 ↓	69.26 ↑
	BALD	59.91 ↑	n.a.*	52.13 ↑	69.17 ↑	44.79 ↑	36.14/36.34 ↑	68.53 ↑	64.24 ↑
	EKM	52.44 ↓	59.62 ↓	47.65 ↓	69.77 ↑	46.49 ↑	32.87/32.75 ↓	61.08 ↓	66.60 ↑
	PE	53.75 ↓	58.65 ↓	54.44 ↑	56.86 ↑	42.82 ↑	37.11/37.20 ↑	66.96 ↓	69.26 ↑

表 2: 不同 LLM 的 ActiveLLM 以及其他主动学习策略 (LC、BALD、EKM 和 PE) 在 GLUE 任务上的 BERT-base 少样本结果 (32 个实例)。¹ ‘Active 200/100’ 描述了未标记实例批次的大小。所有结果均为 5 次运行的平均值 (准确率)。标准差见附录表 5。

* 执行 48 小时后无可用结果。

		QNLI	QQP	RTE	SST2	WNLI	MNLI	MRPC	COLA
	Average	265.26	279.11	11.01	170.34	5.17	627.12	12.31	25.29
AL	LC	81.58	271.40	6.39	53.11	3.55	297.00	5.50	11.07
	BALD	812.26	n.a.*	22.34	520.11	8.37	1610.23	30.54	68.26
	EKM	85.16	289.39	7.23	53.53	4.40	306.43	6.08	10.02
	PE	82.02	276.54	6.47	53.01	4.37	293.24	5.58	10.22

表 3: 表 2 中 GLUE 任务上各主动学习策略 (LC、BALD、EKM 和 PE) 的运行时间。运行时间 (以分钟计) 指在 6 轮主动学习迭代中选取 32 个实例所需的总时间。

* 执行 48 小时后仍无可用结果。

4.5 ActiveLLM and Few-Shot Learning

- Can ActiveLLM improve state-of-the-art few-shot learning methods?

尽管在少样本场景中相比随机采样和其他 AL 策略显示出显著改进, 人们可能会问为何不直接使用少样本学习方法。因此, 我们评估了 ActiveGPT4 单独使用以及与最先进的少样本学习方法 ADAPET、PERFECT 和 SetFit 结合使用的性能。对于每种方法, 我们必须使用有可用实现的不同数据集: ADAPET 使用 CTI, PERFECT 使用 SST-2¹, SetFit 使用 AgNews。该实验的结果如图 5 所示。ActiveGPT4 显著优于少样本学习方法 ADAPET 和 PERFECT (分别提升 +17.64% 和 +24.29%)。虽然单独使用

¹原始实现仅支持 RoBERTa-Large。

时未超越 SetFit, 但 ActiveLLM 与每种少样本学习方法结合后, 均显著提升了所有这些方法的性能。这表明少样本学习方法可以通过实例选择得到改进, 并且在某些情况下, 这种选择甚至比使用少样本学习方法本身更为重要。

4.6 Non-Few-Shot-Scenarios

在最后的实验中, 我们研究了 ActiveLLM 在非少样本场景中的表现, 以及它是否可用于克服其他 AL 策略的冷启动问题。为此, 我们选择每次 AL 迭代的样本量为 25 个示例, 并评估该过程直至 300 个示例。

4.6.1 Iterated Querying

- How does the method perform in non-few-shot scenarios?

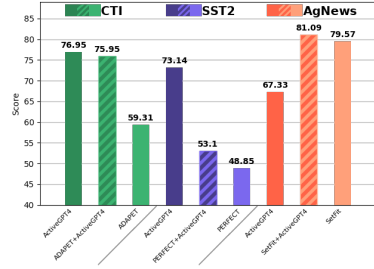


Figure 5: Comparison of ActiveGPT4, both independently and in combination with few-shot methods, across different tasks (32 instances). The methods include ADAPET (evaluated using BERT-base on CTI with F1), PERFECT (using RoBERTa-Large on SST2 with accuracy), and SetFit (using BERT-base on AG-News with accuracy).

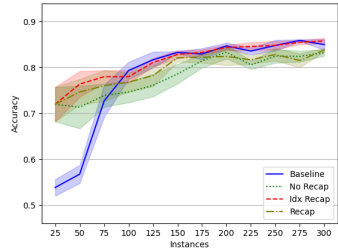


Figure 6: Iterated querying on the SST-2 dataset with different strategies of recalling already labeled instances. All results are averaged over 5 runs (accuracy).

100 instances. Beyond this point, it performs comparably to random sampling. Furthermore, the figure highlights the importance of recalling instances that have already been selected, as the no-recap mode performs the worst. Providing the model with the indices of these instances appears to be the most efficient method, as it does not overly complicate or enlarge the prompt.

These results motivate the subsequent experiment to investigate whether ActiveLLM could be a viable solution to overcome the cold-start problem of learning strategies.

4.6.2 Overcoming the Cold-Start Problem

- Is ActiveLLM capable of overcoming the cold-start problem in other AL strategies?

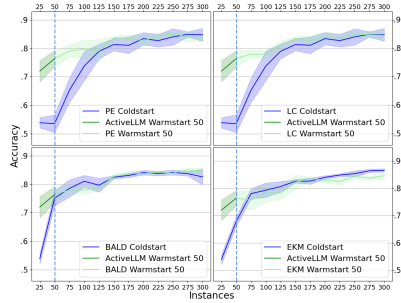


Figure 7: Comparison of BALD, EKM, LC, and PE with and without selected instances by ActiveGPT4 as seed label set on SST-2. All results are averaged over 5 runs (accuracy).

As outlined in Section

Once again, a selection size of 25 was used, and the experiments continued until reaching a total of 300 examples.

5 Discussion & Conclusion

Utilizing instruction-tuned LLMs presents several advantages for AL with BERT-like transformers. Since these LLMs exhibit substantial zero-shot capabilities, they overcome the cold-start problem. This is crucial for low-data regimes and few-shot scenarios, which are becoming increasingly important (Yu and Raschka, 2020). Realized in the ActiveLLM method, we show that this procedure indeed excels in few-shot scenarios, outperforming the baseline and common AL strategies for transformers. Furthermore, in our experiments, ActiveLLM significantly outperforms state-of-the-art few-shot learning methods, ADAPET and PERFECT, while also enhancing the performance of SetFit, ADAPET, and PERFECT when used in combination. We also demonstrate a variant of ActiveLLM with iterative querying, which is applicable to non-few-shot scenarios and incorporates feedback from previous iterations. In a combined setting, we show that ActiveLLM overcomes the cold-start problem inherent in conventional AL strategies.

Moreover, ActiveLLM is particularly suitable for practical scenarios as it effectively eliminates

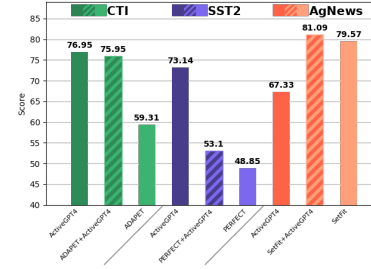


图 5: ActiveGPT4 在不同任务上独立使用以及与少样本方法结合的性能比较 (32 个实例)。方法包括 ADAPET (在 CTI 上使用 BERT-base 评估, 指标为 F1)、PERFECT (在 SST2 上使用 RoBERTa-Large 评估, 指标为准确率) 以及 SetFit (在 AGNews 上使用 BERT-base 评估, 指标为准确率)。

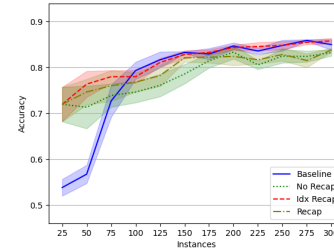


图 6: 在 SST-2 数据集上使用不同策略回忆已标注实例的迭代查询结果。所有结果为 5 次运行的平均值 (准确率)。

图6表明, ActiveGPT4 在实例数量约 100 以内的少样本场景中尤其有益。超过此范围后, 其性能与随机采样相当。此外, 该图强调了回忆已选择实例的重要性, 因为无回忆模式表现最差。为模型提供这些实例的索引似乎是最有效的方法, 因为它不会过度复杂化或扩大提示。

这些结果促使我们进行后续实验, 以探究 ActiveLLM 是否可能成为克服学习策略冷启动问题的可行解决方案。

4.6.2 Overcoming the Cold-Start Problem

- Is ActiveLLM capable of overcoming the cold-start problem in other AL strategies?

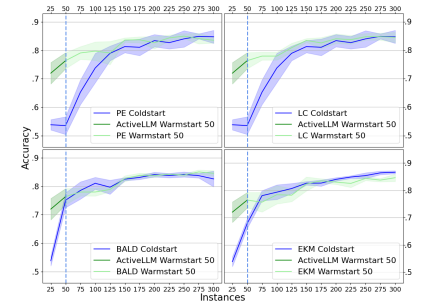


图 7: 在 SST-2 上比较 BALD、EKM、LC 和 PE 方法使用与不使用由 ActiveGPT4 选择的实例作为种子标签集的效果。所有结果为 5 次运行的平均值 (准确率)。

如第

再次使用 25 的选取规模, 实验持续进行直至总共达到 300 个示例。

5 Discussion & Conclusion

利用指令调优的大型语言模型为基于 BERT 类变换器的主动学习带来了若干优势。由于这些大型语言模型展现出强大的零样本能力, 它们克服了冷启动问题。这对于低数据状态和少样本场景至关重要, 这些场景正变得越来越重要 (Yu and Raschka, 2020)。在 ActiveLLM 方法中实现, 我们证明该流程确实在少样本场景中表现出色, 超越了基线以及针对变换器的常见主动学习策略。此外, 在我们的实验中, ActiveLLM 显著优于最先进的少样本学习方法 ADAPET 和 PERFECT, 同时在与 SetFit、ADAPET 和 PERFECT 结合使用时也提升了它们的性能。我们还展示了 ActiveLLM 的一个变体, 该变体采用迭代查询, 适用于非少样本场景, 并融入了先前迭代的反馈。在组合设置中, 我们证明 ActiveLLM 克服了传统主动学习策略固有的冷启动问题。

the dependency between the query and the successor model. In our experiments, the AL methods LC, BALD, EKM, and PE required several hours to complete (see Table 4), whereas even the more efficient AL methods commonly discussed in the literature typically require several minutes per querying iteration (Yuan et al., 2020). While some of the LLMs used in ActiveLLM require significantly more resources, our approach eliminates the need for model training during the AL selection process, as required by many other AL methods. Additionally, only a small portion of the unlabeled data is considered, still resulting in significant improvements while requiring just a few seconds for querying. Furthermore, the current cost-free availability of these LLMs through chat interfaces eliminates the need for extensive resources or financial investment. If cost-free availability persists, it democratizes AL, making it accessible to a wider audience rather than limiting it to hypothetical scenarios that assume unlimited time, resources, or funding.

5.1 LLM-based Active Learning Strategies

The question remains as to how LLMs choose instances. While they clearly explain their approach, including avoiding redundancy, selecting diverse instances, and considering ambiguities (see Figure 10), it is nearly impossible to assess whether the models truly perform as claimed. However, LLMs are known for their ability to identify topics and patterns in texts, which suggests that diversity-based sampling, a common approach in other procedures, is a valid method. Due to this property, they can also identify differences between instances, enabling them to recognize unique, representative, frequent, or highly information-dense examples. Similarly, we expect these models to perform well on the tasks themselves, which is why they could be relatively proficient at selecting a balanced set of all classes. However, these models cannot inspect their own internals, which means they are not performing uncertainty sampling based on these internals. Related to uncertainty, LLMs might be very well suited for identifying difficult or ambiguous examples. Occasionally, the models mention that they try to filter out any instances that are anomalies that would teach the successor model incorrect patterns. In iterated querying mode, the language models often directly mention strategies regarding the feed-

back, explaining that for selecting the instances, they incorporate past instances and try to avoid redundancy, but also track very ambiguous instances from which more data might be needed in the successor training.

5.2 Limitations and Future Work

In this study, we focus on AL for BERT-like transformer models. While LLMs such as GPT-4 offer greater capabilities, they often require substantial computational resources, incur high API costs, and raise concerns regarding data privacy and control (Wang et al., 2024a; Samuel, 2024; Treviso et al., 2023; Bosley et al., 2023). In contrast, BERT-like models remain an important choice for many practical applications due to their efficiency, open availability, and competitive performance when fine-tuned on domain-specific tasks. Although ActiveLLM leverages LLMs, their use is limited to the labeling phase and is not required after training. As such, the resulting BERT-like classifier can be deployed without relying on external APIs or continued access to proprietary LLMs. Nonetheless, it is important to acknowledge that the integration of LLMs during training may still pose privacy challenges with regard to the data exposed to them. Our approach therefore offers a trade-off: it combines the strengths of LLMs for data-efficient training while preserving the lightweight and self-contained nature of traditional transformer classifiers at inference time. It might also be of interest to use the LLM directly as an oracle in our framework, i.e., to have it label the selected instances directly. However, as this study focuses on the selection of training examples, we chose not to introduce additional potential points of failure. Instead, we refer readers to the work of Wang et al. (2024b); Walshe et al. (2025), who explore the use of LLMs in the annotation process. We also look forward to applying ActiveLLM to LLMs themselves. As reported by Zhang et al. (2022b), the selection of training examples for in-context learning can have a significant impact on LLM performance, making ActiveLLM a promising method in this context as well.

While the experiments confirm the strong performance of ActiveLLM’s general design, we look forward to further exploring fine-grained LLM pipeline and prompt optimization techniques, including modular and programmable approaches (Khatab et al., 2023). In particular, the context

此外, ActiveLLM 特别适用于实际场景, 因为它有效消除了查询与后续模型之间的依赖关系。

在我们的实验中, 主动学习方法 LC、BALD、EKM 和 PE 需要数小时才能完成 (见表3), 而文献中常讨论的、更高效的主动学习方法通常也需要每次查询迭代数分钟 (Yuan et al., 2020)。尽管 ActiveLLM 中使用的一些大型语言模型需要显著更多的资源, 但我们的方法消除了了在主动学习选择过程中进行模型训练的需求, 而这是许多其他主动学习方法所必需的。此外, 仅考虑未标记数据的一小部分, 仍能带来显著改进, 而查询仅需几秒钟。再者, 这些大型语言模型通过聊天界面当前可免费获取, 消除了对大量资源或资金投入的需求。如果免费可用性持续下去, 它将使主动学习民主化, 使其更广泛地普及, 而非局限于那些假设拥有无限时间、资源或资金的理想化场景。

问题仍然在于大语言模型如何选择实例。尽管它们清晰地解释了自身方法, 包括避免冗余、选择多样化实例以及考虑歧义性 (见图 10), 但几乎无法评估模型是否真正如其所述那样运作。然而, 大语言模型以其识别文本主题与模式的能力而闻名, 这表明基于多样性的抽样——其他流程中的常见方法——是一种有效策略。凭借这一特性, 它们也能识别实例间的差异, 从而能够辨识出独特、具代表性、频繁出现或信息密度极高的示例。类似地, 我们预期这些模型在任务本身上表现良好, 这解释了它们为何可能相对擅长选择所有类别的平衡集合。然而, 这些模型无法检视其内部机制, 这意味着它们并非基于这些内部机制执行不确定性抽样。与不确定性相关的是, 大语言模型可能非常适合识别困难或模糊的示例。有时, 模型会提及它们试图过滤掉那些可能教导后续模型错误模式的异常实例。在迭代查询模式下, 语言模型常直接提及关于反馈的策略, 解释称在选择实例时, 它们会纳入过往实例并试图避免冗余, 同时追踪那些可能需要更多数据用于后续训练的极具歧义的实例。

5.1 Limitations and Future Work

在本研究中, 我们专注于针对类 BERT 的 Transformer 模型进行主动学习。虽然像 GPT-4 这样的大型语言模型 (LLMs) 提供了更强大的能力, 但它们通常需要大量的计算资源, 产生高昂的 API 调用成本, 并引发关于数据隐私和控制权的担忧 (Wang et al., 2024a; Samuel, 2024; Treviso et al., 2023; Bosley et al., 2023)。相比之下, 类 BERT 模型因其高效性、开源可用性以及在特定领域任务上微调后的竞争性表现, 仍然是许多实际应用中的重要选择。尽管 ActiveLLM 利用了 LLMs, 但其使用仅限于标注阶段, 训练完成后便不再需要。因此, 最终得到的类 BERT 分类器可以在不依赖外部 API 或持续访问专有 LLM 的情况下部署。尽管如此, 必须承认在训练过程中整合 LLMs, 对于暴露给它们的数据而言, 可能仍然会带来隐私挑战。因此, 我们的方法提供了一种权衡: 它结合了 LLMs 在数据高效训练方面的优势, 同时在推理时保留了传统 Transformer 分类器的轻量级和自包含特性。在我们的框架中直接使用 LLM 作为标注源 (即让其直接为选定的实例打标签) 也可能具有研究价值。然而, 由于本研究侧重于训练样本的选择, 我们选择不引入额外的潜在故障点。相反, 我们建议读者参考 Wang et al. (2024b); Walshe et al. (2025) 的工作, 他们探索了 LLMs 在标注过程中的应用。我们也期待将 ActiveLLM 应用于 LLMs 本身。正如 Zhang et al. (2022b) 所报告的, 为上下文学习选择训练样本能对 LLM 性能产生显著影响, 这使得 ActiveLLM 在此背景下也成为一种有前景的方法。

虽然实验证实了 ActiveLLM 通用设计的强大性能, 但我们期待进一步探索细粒度的 LLM 流程和提示优化技术, 包括模块化和可编程的方法 (Khatab et al., 2023)。特别是, 上下文长度对 ActiveLLM 至关重要。如果提示变得过长, LLMs 往往会遗忘任务或无法对多个实例进行推理。尽管有报道称某些模型能够处理长

size is critical for ActiveLLM. If the prompts get too large, the LLMs tend to forget the task or are not able to reason over the various instances. Although some models were reported to be able to handle long context inputs, we found that they were not able to reason over long contexts. This is in line with the findings of Hsieh et al. (2024) who argue that simple retrieval-based tests, like the common needle-in-a-haystack test, are not suitable for testing long context understanding. In our results, we noticed differences for tasks with longer instances on average. We also hypothesize that the length may be the reason why LLMs sometimes omit certain instructions or information, such as the provided labels in our tests of the iterated querying mode. We are interested in considering not only the instances as a length restriction but also the token count in future work. Furthermore, in our experiments we wanted to show that the general prompt optimized in Section 4.2 is applicable to other models and datasets. Another approach would be to consider the prompt as adaptive and tune it for each model on the respective development set. While this introduces a new hyperparameter, we expect it to improve performance in most cases. Furthermore, optimizing the prompt for GPT-4 may also explain why certain types of LLMs perform worse.

A common problem in the field of LLMs is that many established and open benchmarks have leaked into the training data. This could bias the results of our work, as a model may already be familiar with the dataset. While we cannot completely avoid this phenomenon, we have tried to use a less known dataset for prompt engineering, as well as smaller and older models, which are less susceptible to data leakage.

When we describe ActiveLLM as much more efficient and less resource-intensive for practitioners than other AL methods, we assume the availability of cost-free chat interfaces or low-cost APIs. Certainly, the LLMs that ActiveLLM relies on consume considerable resources, even if they are only used for a very short time during sample selection for labeling. In terms of practicality, our experiments also focus on the use of the chat versions of the LLMs. However, directly using the models, e.g., with the APIs, might yield different results. While this might only play an insignificant difference for Llama or Mixtral, some chat interfaces, like those from OpenAI or Google, hide

parameters, preceding prompts, and exact version declarations. We see our results as a baseline that might be improved with different parameters or without a preceding prompt in the future.

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上下文输入，但我们发现它们无法对长上下文进行有效推理。这与Hsieh et al. (2024) 的发现一致，他们认为简单的基于检索的测试（如常见的“大海捞针”测试）并不适合测试长上下文理解能力。在我们的结果中，我们注意到对于平均实例长度较长的任务存在差异。我们还假设，长度可能是 LLMs 有时会忽略某些指令或信息（例如在我们测试迭代查询模式时提供的标签）的原因。我们希望在未来的工作中不仅将实例数量视为长度限制，也考虑令牌数量。此外，在我们的实验中，我们希望证明在第4.2节中优化的通用提示适用于其他模型和数据集。另一种方法是将提示视为自适应的，并针对每个模型在各自的开发集上进行调优。虽然这会引入一个新的超参数，但我们预计在大多数情况下它能提升性能。此外，为 GPT-4 优化提示也可能解释为何某些类型的 LLMs 表现更差。大语言模型领域的一个常见问题是，许多已确立的开放基准已泄露至训练数据中。这可能会使我们的研究结果产生偏差，因为模型可能已对该数据集有所了解。虽然我们无法完全避免这一现象，但我们尝试使用一个较为冷门的数据集进行提示工程，并采用规模较小、版本较旧的模型，这些模型对数据泄露的敏感性较低。

当我们描述 ActiveLLM 相较于其他主动学习方法对实践者而言效率更高且资源密集度更低时，我们假设存在免费的聊天接口或低成本 API 可供使用。诚然，ActiveLLM 所依赖的大语言模型消耗大量资源，即使它们仅在样本选择进行标注的极短时间内被使用。在实际应用层面，我们的实验也聚焦于使用这些大语言模型的聊天版本。然而，直接使用模型（例如通过 API）可能会产生不同的结果。虽然这对于 Llama 或 Mixtral 等模型可能仅带来微不足道的差异，但一些聊天接口（如来自 OpenAI 或 Google 的）会隐藏参数、前置提示词和确切的版本声明。我们将自己的结果视为一个基线，未来或许可以通过调整不同参数或移除前置提示词来加以改进。

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Appendix

A Full Prompt Design for Iterated Querying Mode

图 9 展示了 ActiveLLM 第二种变体的综合提示设计。与第一种变体的提示设计相比，此设计包含了“无回顾”、“回顾”和“索引回顾”参数。在“无回顾”和“回顾”场景中，呈现给模型的未标注实例是紧接在最后一个已标注实例之后的那些实例，其范围延伸至最后一个已标注实例的索引加上批次大小（通常为 200 个实例）。相比之下，“索引回顾”场景则保留最后呈现的实例，但包含所有先前的实例，允许模型访问作为已标注给出的特定索引处的文本。此外，图 9 显示了“建议”参数，该参数概述了各种主动学习策略，包括代表性、多样性、难度、分层、平衡、时空分布以及偏差避免。

B Specialized CTI Dataset Details

Split	Count	Relevant	Not Relevant
Train (full)	1800	949	851
Train	32	16	16
Dev (full)	600	273	327
Dev	8	4	4
Test	601	304	297
Total	3001	1526	1475

表 4: CTI 数据集的划分，显示数据集中相关与不相关标签的数量。

专业化 CTI 少样本数据集由推文组成，这些推文根据其在 2021 年 Microsoft Exchange Server 数据泄露事件中对网络安全专家的相关性进行了二元标注。该数据集被划分为完整训练集和少样本训练集与开发集，在我们的案例中仅完整训练集相关，并从中采样了 32 个实例。划分（训练集、开发集）分别包含完整集的 1800 和 600 个实例，以及少样本集的 32 和 8 个实例。测试集在两种情况下相同，均由 601 个实例组成。详细信息亦在表 4 中给出。

C LLM Details

对于大语言模型 GPT-4²、GPT-4o³、o1⁴、GPT-3.5⁵、Gemini-Ultra⁶ 以及 Mistral Large⁷，我们使用了由 OpenAI、Google 和 Mistral 提供的原生聊天界面。对于 Llama 3 70B⁸ 和 Mixtral 8x7B⁹，我们使用了 deepinfra 提供的聊天界面。我们未使用额外的系统提示，最大回答长度为 2048 个令牌，温度为 0.7，Top P 为 0.9，Top K 设置为 0。对于所有交互，我们开启了新的聊天会话并禁用了记忆功能。

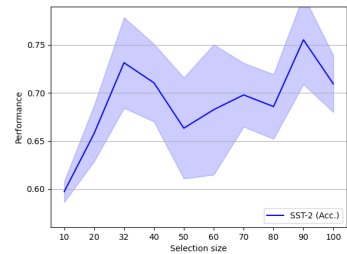


图 8: ActiveGPT4 选择的不同示例数量变化。所有结果均为在 SST-2（准确率）数据集上 5 次运行的平均值。

²<https://chatgpt.com/?model=gpt-4> - 于 2024 年 1 月查询
³<https://chatgpt.com/?model=gpt-4o> - 于 2024 年 5 月查询
⁴<https://chatgpt.com/?model=o1> - 于 2025 年 1 月查询
⁵<https://chatgpt.com/?model=text-davinci-002-render-sha> - 于 2024 年 4 月查询
⁶<https://gemini.google.com/app> -> Gemini-Ultra - 于 2024 年 4 月查询
⁷<https://chat.mistral.ai/chat> -> 模型: Large - 于 2024 年 4 月查询
⁸<https://deepinfra.com/meta-llama/Meta-Llama-3-70B-Instruct> - 于 2024 年 5 月查询
⁹<https://deepinfra.com/mistralai/Mixtral-8x7B-Instruct-v0.1> - 于 2024 年 4 月查询

Appendix

A Full Prompt Design for Iterated Querying Mode

Figure 9 illustrates the comprehensive prompt design for the second variant of ActiveLLM. Compared to the prompt design of the first variant, this one contains the parameters 'No Recap', 'Recap' and 'Index Recap'. In scenarios 'No Recap' and 'Recap', the unlabeled instances presented to the model are the ones immediately following the last labeled instance, extending up to the index of this last labeled instance plus the batch size, typically 200 instances. In contrast, the 'Index Recap' scenario maintains the last presented instance but includes all preceding instances, allowing the model to access texts from the specific indices given as labeled. Additionally, Figure 9 shows the 'advice' parameter, which outlines various AL strategies, including representativeness, diversity, difficulty, stratification, balance, temporal/special distribution, and bias avoidance.

B Specialized CTI Dataset Details

Split	Count	Relevant	Not Relevant
Train (full)	1800	949	851
Train	32	16	16
Dev (full)	600	273	327
Dev	8	4	4
Test	601	304	297
Total	3001	1526	1475

Table 5: Split of the CTI dataset with count of relevant and not relevant labels in the datasets.

The Spezialised CTI few-shot dataset consists of tweets that are binary labeled based on their relevance to cybersecurity experts during the Microsoft Exchange Server data breach in 2021. The dataset is split into a full and a few-shot training and development set, with only the full training set being relevant in our case, from which 32 instances are sampled. The splits (train, dev) consist of 1800 and 600 instances for the full set and 32 and 8 instances for the few-shot set, respectively. The test set is the same in both cases and consists of 601 instances. The details are also given in Table 5.

C LLM Details

For the LLMs GPT-4², GPT-4o³, o1⁴, GPT-3.5⁵, Gemini-Ultra⁶, and Mistral Large⁷, we utilized the native chat interfaces provided by OpenAI, Google, and Mistral. For Llama 3 70B⁸ and Mixtral 8x7B⁹, we used the chat interface provided by deepinfra. We used no extra system prompts, a maximum answer length of 2048 tokens, a temperature of 0.7, Top P of 0.9, and Top K set to 0. For all interactions, we opened a new chat session and disabled memory functions.

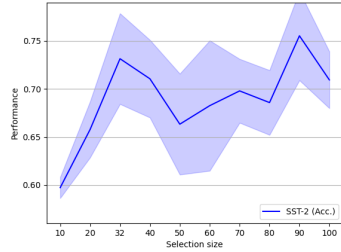


Figure 8: Varying selection size of examples to be selected by ActiveGPT4. All results are averaged over 5 runs on the SST-2 (accuracy) dataset.

²<https://chatgpt.com/?model=gpt-4> - queried in January 2024
³<https://chatgpt.com/?model=gpt-4o> - queried in May 2024
⁴<https://chatgpt.com/?model=o1> - queried in January 2025
⁵<https://chatgpt.com/?model=text-davinci-002-render-sha> - queried in April 2024
⁶<https://gemini.google.com/app> -> Gemini-Ultra - queried in April 2024
⁷<https://chat.mistral.ai/chat> -> Model: Large - queried in April 2024
⁸<https://deepinfra.com/meta-llama/Meta-Llama-3-70B-Instruct> - queried in May 2024
⁹<https://deepinfra.com/mistralai/Mixtral-8x7B-Instruct-v0.1> - queried in April 2024

Consider yourself in the position of an active learning component to help a human annotator. You have to choose the instances that the annotator has to label. You are given (the guidelines for the task and) a set of instances (of a dataset) (that were already labeled and afterwards a set of instances of the dataset) (of a dataset and the indices of the instances that were already labeled). You can only choose (32) instances. (You would ideally want to choose those instances that would provide the most informative and diverse data for the model. Here are some strategies to consider: advice)

(Label Guidelines for the human annotator: guidelines)

(The output format should be a list of the instances that you would label, separated by a comma.) (Please think step by step about what you would do to select the instances to label. After this provide the list of instances that you would label, separated by a comma.) (Please describe instance by instance why you would select or not select it for labelling. Do not stop before you successfully found (32) that you would suggest to label. After this provide the list of the instances that you would label, separated by a comma.) For example, if you would label the instances 1, 4, 5 then the output should be: 1, 4, 5.

(The following instances have already been labeled (separated with "\n #### \n")):

(The following instances are given to you (separated with "\n #### \n")):

(The following instances have already been labeled (indices of the dataset above)):

advice: You would ideally want to choose those instances that would provide the most informative and diverse data for the model. Here are some strategies to consider:

Representativeness: Select instances that are representative of the entire dataset. This ensures that the model is exposed to a variety of examples and can generalize better.

Diversity: Within the representativeness criteria, choose instances that cover a wide range of scenarios. This could include edge cases or less common situations.

Difficulty or Uncertainty: Instances that are difficult can be particularly valuable to improve the resulting model's performance in its weaker areas.

Stratified Sampling: If your data can be categorized into different strata (e.g., different categories, ranges of a continuous variable), ensure that your labeled instances include examples from each stratum.

Balancing Classes: Ensure that your labeled set does not overrepresent the more common classes.

Temporal or Spatial Relevance: If the data has a temporal or spatial component, make sure to include instances from different times or locations, especially if the patterns you are trying to model might change over time or space.

Avoid Bias: Be mindful of not introducing biases with your selection. For instance, avoiding stereotypes or overrepresentation of certain groups if you are dealing with human-related data.

Guidelines
No/Recap
Index Recap
Instances
Advice
Guidelines
No CoT
Normal CoT
Explanation
Instances
Recap
Index Recap

图 9: ActiveLLM 迭代查询模式的完整提示设计, 包括'advice' 参数的文本, 该参数描述了几种主动学习策略。

		QNLI	QQP	RTE	SST2	WNLI	MNLI-(m/mm)	MRPC	COLA
200	Baseline	11.10	1.08	1.31	2.09	3.54	2.43/3.23	2.46	8.32
	GPT-4	9.53	2.02	3.61	5.28	1.61	2.20/2.77	2.48	3.45
	GPT-4o	10.70	0.60	1.90	5.28	2.89	0.33/0.16	1.89	4.09
	o1	8.52	2.41	2.25	2.06	2.82	2.18/1.23	0.49	2.25
	Mistral Large	10.77	1.85	2.41	4.46	1.61	1.69/1.23	1.42	1.87
100	Llama 3 70B	10.34	1.64	2.06	1.87	1.89	0.86/1.04	2.87	2.61
	GPT-3.5	11.10	1.66	1.99	5.43	5.33	2.93/3.27	2.37	2.26
	Mistral Large	9.11	1.42	1.96	4.48	2.23	2.55/3.08	0.80	1.34
	Gemini-Ultra	11.65	3.68	1.34	0.00	1.61	2.40/2.79	0.62	0.24
	Mixtral 8x7B	4.24	2.05	1.29	2.55	3.67	2.26/3.18	0.55	1.74
AL	LC	12.10	1.43	1.81	2.28	2.14	3.18/3.38	2.67	1.41
	BALD	9.05	n.a.	0.66	1.59	3.21	3.23/3.75	0.13	6.97
	EKM	4.90	4.59	0.96	2.35	1.00	0.66/0.70	5.35	3.63
	PE	9.20	1.43	1.81	2.28	2.14	2.29/2.65	2.67	1.41

表 5: 表 2 中结果的标准差。

Consider yourself in the position of an active learning component to help a human annotator. You have to choose the instances that the annotator has to label. You are given (the guidelines for the task and) a set of instances (of a dataset)(that were already labeled and afterwards a set of instances of the dataset)(of a dataset and the indices of the instances that were already labeled). You can only choose (32) instances. (You would ideally want to choose those instances that would provide the most informative and diverse data for the model. Here are some strategies to consider: advices)	Guidelines No/Recap Index Recap Instances Advice
(Label Guidelines for the human annotator: guidelines)	Guidelines
(The output format should be a list of the instances that you would label, separated by a comma.) (Please think step by step about what you would do to select the instances to label. After this provide the list of instances that you would label, separated by a comma.) (Please describe instance by instance why you would select or not select it for labelling. Do not stop before you successfully found (32) that you would suggest to label. After this provide the list of the instances that you would label, separated by a comma.) For example, if you would label the instances 1, 4, 5 then the output should be: 1, 4, 5.	No CoT Normal CoT
(The following instances have already been labeled (separated with "\n #### \n")):	Explanation Instances
The following instances are given to you (separated with "\n #### \n"):	Recap
(The following instances have already been labeled (indices of the dataset above)):	Index Recap
advices: You would ideally want to choose those instances that would provide the most informative and diverse data for the model. Here are some strategies to consider:	
Representativeness: Select instances that are representative of the entire dataset. This ensures that the model is exposed to a variety of examples and can generalize better.	
Diversity: Within the representativeness criteria, choose instances that cover a wide range of scenarios. This could include edge cases or less common situations.	
Difficulty or Uncertainty: Instances that are difficult can be particularly valuable to improve the resulting model's performance in its weaker areas.	
Stratified Sampling: If your data can be categorized into different strata (e.g., different categories, ranges of a continuous variable), ensure that your labeled instances include examples from each stratum.	
Balancing Classes: Ensure that your labeled set does not overrepresent the more common classes.	
Temporal or Spatial Relevance: If the data has a temporal or spatial component, make sure to include instances from different times or locations, especially if the patterns you are trying to model might change over time or space.	
Avoid Bias: Be mindful of not introducing biases with your selection. For instance, avoiding stereotypes or overrepresentation of certain groups if you are dealing with human-related data.	

Figure 9: Complete prompt design for the iterated querying mode of ActiveLLM, including the text of the ‘advice’ parameter, which represents the description of several AL strategies.

		QNLI	QQP	RTE	SST2	WNLI	MNLI-(m/mm)	MRPC	COLA
200	Baseline	11.10	1.08	1.31	2.09	3.54	2.43/3.23	2.46	8.32
	GPT-4	9.53	2.02	3.61	5.28	1.61	2.20/2.77	2.48	3.45
	GPT-4o	10.70	0.60	1.90	5.28	2.89	0.33/0.16	1.89	4.09
	o1	8.52	2.41	2.25	2.06	2.82	2.18/1.23	0.49	2.25
	Mistral Large	10.77	1.85	2.41	4.46	1.61	1.69/1.23	1.42	1.87
100	Llama 3 70B	10.34	1.64	2.06	1.87	1.89	0.86/1.04	2.87	2.61
	GPT-3.5	11.10	1.66	1.99	5.43	5.33	2.93/3.27	2.37	2.26
	Mistral Large	9.11	1.42	1.96	4.48	2.23	2.55/3.08	0.80	1.34
	Gemini-Ultra	11.65	3.68	1.34	0.00	1.61	2.40/2.79	0.62	0.24
	Mixtral 8x7B	4.24	2.05	1.29	2.55	3.67	2.26/3.18	0.55	1.74
AL	LC	12.10	1.43	1.81	2.28	2.14	3.18/3.38	2.67	1.41
	BALD	9.05	n.a.	0.66	1.59	3.21	3.23/3.75	0.13	6.97
	EKM	4.90	4.59	0.96	2.35	1.00	0.66/0.70	5.35	3.63
	PE	9.20	1.43	1.81	2.28	2.14	2.29/2.65	2.67	1.41

Table 6: Standard deviations of the results in Table 3.

		QNLI	QQP	RTE	SST2	WNLI	MNLI-(m/mm)	MRPC	COLA
200	基线	60.46	66.50	50.69	58.39	33.24	33.76/33.32	73.63	69.03
	GPT-4	67.63 ↑	65.30 ↓	54.44 ↑	82.00 ↑	42.54 ↑	34.46/34.14 ↑	70.39 ↓	69.20 ↑
	GPT-4o	65.54 ↑	63.57 ↓	53.21 ↑	68.78 ↑	43.10 ↑	32.49/32.99 ↓	74.46 ↑	69.20 ↑
	o1	57.57 ↓	67.71 ↑	58.63 ↑	68.65 ↑	42.54 ↑	34.45/33.78 ↑	68.38 ↓	69.19 ↑
	Mistral Large	58.98 ↓	66.63 ↑	57.33 ↑	61.56 ↑	54.37 ↑	35.59/35.33 ↑	69.12 ↓	69.19 ↑
100	Llama 3 70B	58.44 ↓	65.46 ↓	53.14 ↑	77.39 ↑	43.38 ↑	36.03/35.51 ↑	68.63 ↓	69.34 ↑
	GPT-3.5	60.47 ↑	67.21 ↑	55.81 ↑	68.39 ↑	58.31 ↑	35.56/36.22 ↑	72.01 ↓	69.13 ↑
	Mistral Large	59.36 ↓	66.91 ↑	49.96 ↓	79.66 ↑	50.70 ↑	36.56/36.03 ↑	68.38 ↓	69.20 ↑
	Gemini-Ultra	60.30 ↓	66.23 ↓	54.37 ↑	49.08 ↓	40.00 ↑	35.55/36.20 ↑	69.95 ↓	69.38 ↑
	Mixtral 8x7B	61.15 ↑	66.30 ↓	58.77 ↑	71.97 ↑	42.25 ↑	33.31/33.46 ↓	67.70 ↓	69.12 ↑

表 6: RoBERTa: ActiveLLM 在不同 LLM 上于 GLUE 任务的小样本（32 个实例）结果。‘Active 200/100’ 描述了未标注实例批次的大小。所有结果为 5 次运行的平均值（准确率）。

Consider yourself in the position of an active learning component to help a human annotator. [...] The following instances are given to you (seperated with "\n #### \n"):	1. **Understand the Dataset** : The dataset seems to consist of movie-related texts, possibly reviews or descriptions. Understanding the content helps in making informed decisions about which instances to label.
Table 2	Criteria
[...]	
20. dramatically lackluster . [ID: 9108] Then Selected Instance (Figure 10)	2. **Review Already Labeled Data** : Check the characteristics of instances already labeled (0, 1, 5, 6, 12, 17, 23, 24, 31, 34, 41, 47, 55, 61, 67, 73, 80, 90, 97, 100, 106, 115, 123, 131, 139) to avoid redundancy and ensure that new selections add value.
##### 21. been cobbled together onscreen [ID: 12674] ##### 22. blue crush follows the formula , but throws in too many conflicts to keep the story compelling . [ID: 8372] ##### 23. the somber pacing and lack of dramatic fireworks make green dragon seem more like medicine than entertainment . [ID: 36539] ##### 24. in spite of clearly evident poverty and hardship [ID: 55756] ##### 25. wonderfully speculative [ID: 9435] ##### 26. are woven together skilfully [ID: 56634] Then Selected Instance (Figure 10)	Criteria Index Recap
The following instances have already been labeled (indices of the dataset above): 0, 1, 5, 6, 12, 17, 23, 24, 31, 34, 41, 47, 55, 61, 67, 73, 80, 90, 97, 100, 106, 115, 123, 131, 139.	3. **Diversity and Coverage** : Aim to select instances that are diverse in terms of sentiment, length, complexity, and subject matter to ensure broad coverage of the dataset's features.
Index Recap	4. **Avoid Similar Instances** : Avoid selecting instances that are too similar to already labeled ones to maximize the variety of the labeled data.
	5. **Consider Edge Cases and Ambiguites** : Include some instances that represent edge cases or ambiguities, as they can be crucial for training a more accurate and nuanced model.
	Now, I will select 25 instances from the dataset based on these criteria. Let's proceed with the selection: Based on the criteria for diversity, representativeness, and avoiding redundancy with already labeled instances, the selected 25 instances to label are: 2, 8, 14, 20, 26, 32, 38, 44, 50, 56, 64, 70, 76, 85, 91, 98, 104, 110, 119, 127, 135, 142, 148, 154, 160
	All Selected Instances
	These instances were chosen to encompass a range of sentiments (positive, negative, neutral, mixed), diverse content related to movies (different genres, aspects, and perspectives), and include some edge cases or ambiguous instances for robust model training.

图 10: 由 ActiveLLM 在迭代查询模式下生成的 SST-2 示例提示，无指导原则，无建议。（左） GPT-4 对左侧所示提示的回复。（右）

		QNLI	QQP	RTE	SST2	WNLI	MNLI-(m/mm)	MRPC	COLA
	Baseline	60.46	66.50	50.69	58.39	33.24	33.76/33.32	73.63	69.03
200	GPT-4	67.63 ↑	65.30 ↓	54.44 ↑	82.00 ↑	42.54 ↑	34.46/34.14 ↑	70.39 ↓	69.20 ↑
	GPT-4o	65.54 ↓	63.57 ↓	53.21 ↑	68.78 ↑	43.10 ↑	32.49/32.99 ↓	74.46 ↑	69.20 ↑
	o1	57.57 ↓	67.71 ↑	58.63 ↑	68.65 ↑	42.54 ↑	34.45/33.78 ↑	68.38 ↓	69.19 ↑
	Mistral Large	58.98 ↓	66.63 ↑	57.33 ↑	61.56 ↑	54.37 ↑	35.59/35.33 ↑	69.12 ↓	69.19 ↑
100	Llama 3 70B	58.44 ↓	65.46 ↓	53.14 ↑	77.39 ↑	43.38 ↑	36.03/35.51 ↑	68.63 ↓	69.34 ↑
	GPT-3.5	60.47 ↑	67.21 ↑	55.81 ↑	68.39 ↑	58.31 ↑	35.56/36.22 ↑	72.01 ↓	69.13 ↑
	Mistral Large	59.36 ↓	66.91 ↑	49.96 ↓	79.66 ↑	50.70 ↑	36.56/36.03 ↑	68.38 ↓	69.20 ↑
	Gemini-Ultra	60.30 ↓	66.23 ↓	54.37 ↑	49.08 ↓	40.00 ↑	35.55/36.20 ↑	69.95 ↓	69.38 ↑
	Mixtral 8x7B	61.15 ↑	66.30 ↓	58.77 ↑	71.97 ↑	42.25 ↑	33.31/33.46 ↓	67.70 ↓	69.12 ↑

Table 7: RoBERTa: Few-shot results (32 instances) of ActiveLLM with different LLMs on GLUE tasks. ‘Active 200/100’ describes the size of the batch of unlabeled instances. All results are averaged over 5 runs (accuracy).

Consider yourself in the position of an active learning component to help a human annotator. [...] The following instances are given to you (seperated with "\n #### \n"):

Table 2

...

20. dramatically lackluster . [ID: 9108]

Then Selected Instance (Figure 10)

####

21. been cobbled together onscreen [ID: 12674]

####

22. blue crush follows the formula , but throws in too many conflicts to keep the story compelling . [ID: 8372]

####

23. the somber pacing and lack of dramatic fireworks make green dragon seem more like medicine than entertainment . [ID: 36539]

####

24. In spite of clearly evident poverty and hardship [ID: 55756]

####

25. wonderfully speculative [ID: 9435]

####

26. are woven together skillfully [ID: 56634]

Then Selected Instance (Figure 10)

...

The following instances have already been labeled (indices of the dataset above): 0, 1, 5, 6, 12, 17, 23, 24, 31, 34, 41, 47, 55, 61, 67, 73, 80, 90, 97, 100, 106, 115, 123, 131, 139

Index Recap

To select the 25 instances for labeling from the given dataset, I would follow a strategic approach focusing on diversity and representativeness. The goal is to choose instances that cover a wide range of the dataset's characteristics, which would help in building a more robust and accurate model. Here's my step-by-step approach:

1. **Understand the Dataset** The dataset seems to consist of movie-related texts, possibly reviews or descriptions. Understanding the content helps in making informed decisions about which instances to label.

2. **Review Already Labeled Data** Check the characteristics of instances already labeled (0, 1, 5, 6, 12, 17, 23, 24, 31, 34, 41, 47, 55, 61, 67, 73, 80, 90, 97, 100, 106, 115, 123, 131, 139) to avoid redundancy and ensure that new selections add value.

3. **Diversity and Coverage** Aim to select instances that are diverse in terms of sentiment, length, complexity, and subject matter to ensure broad coverage of the dataset's features.

4. **Avoid Similar Instances** Avoid selecting instances that are too similar to already labeled ones to maximize the variety of the labeled data.

5. **Consider Edge Cases and Ambiguities** Include some instances that represent edge cases or ambiguities, as they can be crucial for training a more accurate and nuanced model.

Now, I will select 25 instances from the dataset based on these criteria. Let's proceed with the selection:

Based on the criteria for diversity, representativeness, and avoiding redundancy with already labeled instances, the selected 25 instances to label are:

2, 8, 14, 20, 26, 32, 38, 44, 50, 56, 64, 70, 76, 85, 91, 98, 104, 110, 119, 127, 135, 142, 148, 154, 160

All Selected Instances

These instances were chosen to encompass a range of sentiments (positive, negative, neutral, mixed), diverse content related to movies (different genres, aspects, and perspectives), and include some edge cases or ambiguous instances for robust model training.

Figure 10: Example prompt for SST-2 generated by ActiveLLM in iterated querying mode, no guidelines, and no advice. (left)
GPT-4 response to the prompt shown on the left. (right)